

[Home](#) / [Class - 8 Concise Mathematics Selina](#) / [Algebraic Expressions](#)

Chapter 11

Algebraic Expressions

Class - 8 Concise Mathematics Selina



STUDY MATERIAL ▾

LOGIN

JOIN NOW

Search by lesson title

Chapter 10
Direct and Inverse Variations

Chapter 11
Algebraic Expressions

Chapter 12
Identities

Chapter 13
Factorisation

Chapter 14
Linear Equations in One Variable

Chapter 15
Linear Inequations

Chapter 16
Understanding Shapes

Chapter 17
Special Types of Quadrilaterals

Chapter 18
Constructions

Social Sciences

TV & Video Equipment

Computer

CONTENTS

[Exercise 11\(A\)](#)[Exercise 11\(B\)](#)[Exercise 11\(C\)](#)[Test Yourself](#)

Exercise 11(A)

Question 1(i)

The sum of $a - b + ab$, $b - c + bc$ and $c - a + ca$ is:

1. 0
2. $2(a + b + c)$
3. $ab + bc + ca$
4. none of these



Answer

The sum of $a - b + ab$, $b - c + bc$ and $c - a + ca$ is,

$$= a - b + ab + b - c + bc + c - a + ca$$

$$= (a - a) + (-b + b) + ab + bc + ca.$$

$$= ab + bc + ca$$

Hence, option 3 is the correct option.

Question 1(ii)

$(x^3 - 5x^2 + 7) + (3x^2 + 5x - 2) + (2x^3 - x + 7)$ is equal to:

1. $3x^3 - 2x^2 + 4x + 12$

2. $3x^3 + 2x^2 - 4x + 12$

3. $3x^3 - 2x^2 - 4x - 12$

4. $3x^3 + 2x^2 + 4x + 12$

Answer

$$(x^3 - 5x^2 + 7) + (3x^2 + 5x - 2) + (2x^3 - x + 7)$$

$$= (x^3 + 2x^3) + (-5x^2 + 3x^2) + (5x - x) + (7 - 2 + 7)$$

$$= 3x^3 - 2x^2 + 4x + 12$$

Hence, option 1 is the correct option.

Question 1(iii)

$(x^3 - 5x^2 + 3x + 2) - (6x^2 - 4x^3 + 3x + 5)$ is equal to:

1. $5x^3 + 11x^2 - 2$

2. $5x^3 - 11x^2 + 3$

3. $5x^3 - 11x^2 - 3$

4. $5x^3 + 11x^2 + 3$

Answer

$$\begin{aligned} & (x^3 - 5x^2 + 3x + 2) - (6x^2 - 4x^3 + 3x + 5) \\ &= x^3 - 5x^2 + 3x + 2 - 6x^2 + 4x^3 - 3x - 5 \\ &= (x^3 + 4x^3) + (-5x^2 - 6x^2) + (3x - 3x) + (2 - 5) \\ &= 5x^3 - 11x^2 - 3 \end{aligned}$$

Hence, option 3 is the correct option.

Question 1(iv)

$p - (p - q) - q - (q - p)$ is equal to:

1. $q - p$

2. $p - q$

3. $p + q$

4. $2p - q$

Answer

$$p - (p - q) - q - (q - p)$$

$$= p - p + q - q - q + p$$

$$= (p - p + p) + (q - q - q)$$

$$= p - q$$

Hence, option 2 is the correct option.

Question 1(v)

$(ab - bc) - (ca - bc) + (ca - ab)$ is:

1. $bc - ab$

2. $ab - ca$

3. $2(ab - bc - ca)$

4. 0

Answer

$$(ab - bc) - (ca - bc) + (ca - ab)$$

$$= ab - bc - ca + bc + ca - ab$$

$$= (ab - ab) + (bc - bc) + (-ca + ca)$$

$$= 0$$

Hence, option 4 is the correct option.

Question 2

Separate the constants and variables from the following:

- 7, $7 + x$, $7x + yz$, $\sqrt{5}$, $\sqrt{xy}$, $\frac{3yz}{8}$, $4.5y - 3x$, $8 - 5$, $8 - 5x$, $8x - 5y \times p$ and $3y^2z \div 4x$

Answer

Constants = -7, $\sqrt{5}$ and $8 - 5$

Variables = $7 + x$, $7x + yz$, $\sqrt{xy}$, $\frac{3yz}{8}$, $4.5y - 3x$, $8 - 5x$, $8x - 5y \times p$ and $3y^2z \div 4x$

Question 3

Write the number of terms in each of the following polynomials:

(i) $5x^2 + 3 \times ax$

(ii) $ax \div 4 - 7$

(iii) $ax - by + y \times z$

(iv) $23 + a \times b \div 2$

Answer

(i) Terms are $5x^2$ and $3 \times ax$

Hence, number of terms = 2.

(ii) Terms are $ax \div 4$ and 7

Hence, number of terms = 2.

(iii) Terms are ax , by and $y \times x$

Hence, number of terms = 3.

(iv) Terms are 23 and $a \times b \div 2$

Hence, number of terms = 2.

Question 4

Separate monomials, binomials, trinomials and multinomial from the following algebraic expressions:

$8 - 3x$, xy^2 , $3y^2 - 5y + 8$, $9x - 3x^2 + 15x^3 - 7$, $3x \times 5y$, $3x \div 5y$, $2y \div 7 + 3x - 7$ and $4 - ax^2 + bx + y$

Answer

Monomials are those algebraic expression which consist of one term.

xy^2 , $3x \times 5y$ and $3x \div 5y$ are monomials.

Binomials are those algebraic expression which consist of two terms.

$8 - 3x$ is binomial.

Trinomials are those algebraic expression which consist of three terms.

$3y^2 - 5y + 8$ and $2y \div 7 + 3x - 7$ are trinomials

Multinomial are those algebraic expression which consist two or more terms.

$8 - 3x$, $3y^2 - 5y + 8$, $9x - 3x^2 + 15x^3 - 7$, $2y \div 7 + 3x - 7$ and $4 - ax^2 + bx + y$ are multinomials.

Question 5

Write the degree of each polynomial given below:

(i) $xy + 7z$

(ii) $x^2 - 6x^3 + 8$

(iii) $y - 6y^2 + 5y^8$

(iv) $xyz - 3$

(v) $xy + yz^2 - zx^3$

(vi) $x^5y^7 - 8x^3y^8 + 10x^4y^4z^4$

Answer

(i) Degree = 2

Reason — As the polynomial contains 3 variables, we will find the sum of powers of each term.

The sum of powers of the term $xy = 1 + 1 = 2$

The sum of powers of the term $7z = 1$

$\therefore$ Highest sum of powers = 2

$\therefore$ **Degree of polynomial = 2**

(ii) Degree = 3

Reason — As the polynomial contains 1 variable, degree of a polynomial is the highest power of the variable in a polynomial expression.

Highest power of polynomial = 3

$\therefore$ Degree of polynomial = 3

(iii) Degree = 8

Reason — As the polynomial contains 1 variable, degree of a polynomial is the highest power of the variable in a polynomial expression.

Highest power of polynomial = 8

$\therefore$ Degree of polynomial = 8

(iv) Degree = 3

Reason — As the polynomial contains 3 variables, we will find the sum of powers of each term.

The sum of powers of the term $xyz = 1 + 1 + 1 = 3$

The powers of the term $3 = 0$

$\therefore$ Highest sum of powers = 3

$\therefore$ Degree of polynomial = 3

(v) Degree = 4

Reason — As the polynomial contains 3 variables, we will find the sum of powers of each term.

The sum of powers of the term $xy = 1 + 1 = 2$

The sum of powers of the term $yz^2 = 1 + 2 = 3$

The sum of powers of the term $zx^3 = 1 + 3 = 4$

$\therefore$ Highest sum of powers = 4

$\therefore$ Degree of polynomial = 4

(vi) Degree = 12

Reason — As the polynomial contains 3 variables, we will find the sum of powers of each term.

The sum of powers of the term $x^5y^7 = 5 + 7 = 12$

The sum of powers of the term $-8x^3y^8 = 3 + 8 = 11$

The sum of powers of the term $10x^4y^4z^4 = 4 + 4 + 4 = 12$

$\therefore$ Highest sum of powers = 12

$\therefore$ **Degree of polynomial = 12**

Question 6(i)

Write the coefficient of:

ab in $7abx$

Answer

Coefficient of ab = $7x$.

Question 6(ii)

Write the coefficient of:

$7a$ in $7abx$

Answer

Coefficient of $7a = bx$.

Question 6(iii)

Write the coefficient of:

$5x^2$ in $5x^2 - 5x$

Answer

Coefficient of $5x^2 = 1$.

Question 6(iv)

Write the coefficient of:

8 in $a^2 - 8ax + a$

Answer

Coefficient of 8 = $-ax$.

Question 6(v)

Write the coefficient of:

$4xy$ in $x^2 - 4xy + y^2$

Answer

Coefficient of $4xy = -1$.

Question 7(i)

Evaluate:

$-7x^2 + 18x^2 + 3x^2 - 5x^2$

Answer

$$-7x^2 + 18x^2 + 3x^2 - 5x^2$$

Arrange the polynomial with like terms together and combine them.

$$= (-7x^2 + 18x^2) + 3x^2 - 5x^2$$

$$= (11x^2 + 3x^2) - 5x^2$$

$$= (14x^2 - 5x^2)$$

$$= 9x^2$$

$$\text{Hence, } -7x^2 + 18x^2 + 3x^2 - 5x^2 = 9x^2$$

Question 7(ii)

Evaluate:

$$b^2y - 9b^2y + 2b^2y - 5b^2y$$

Answer

$$b^2y - 9b^2y + 2b^2y - 5b^2y$$

Arrange the polynomial with like terms together and combine them.

$$= (b^2y - 9b^2y) + 2b^2y - 5b^2y$$

$$= (-8b^2y + 2b^2y) - 5b^2y$$

$$= (-6b^2y - 5b^2y)$$

$$= -11b^2y$$

Hence, $b^2y - 9b^2y + 2b^2y - 5b^2y = -11b^2y$

Question 7(iii)

Evaluate:

$$abx - 15abx - 10abx + 32abx$$

Answer

$$abx - 15abx - 10abx + 32abx$$

Arrange the polynomial with like terms together and combine them.

$$= (abx - 15abx) - 10abx + 32abx$$

$$= (-14abx - 10abx) + 32abx$$

$$= (-24abx + 32abx)$$

$$= 8abx$$

Hence, $abx - 15abx - 10abx + 32abx = 8abx$

Question 7(iv)

Evaluate:

$$7x - 9y + 3 - 3x - 5y + 8$$

Answer

$$7x - 9y + 3 - 3x - 5y + 8$$

Arrange the polynomial with like terms together and combine them.

$$(7x - 3x) - 9y + 3 - 5y + 8$$

$$= 4x + (-9y - 5y) + 3 + 8$$

$$= 4x - 14y + (3 + 8)$$

$$= 4x - 14y + 11$$

$$\text{Hence, } 7x - 9y + 3 - 3x - 5y + 8 = 4x - 14y + 11$$

Question 7(v)

Evaluate:

$$3x^2 + 5xy - 4y^2 + x^2 - 8xy - 5y^2$$

Answer

$$3x^2 + 5xy - 4y^2 + x^2 - 8xy - 5y^2$$

Arrange the polynomial with like terms together and combine them.

$$= (3x^2 + x^2) + 5xy - 4y^2 - 8xy - 5y^2$$

$$= 4x^2 + (5xy - 8xy) - 4y^2 - 5y^2$$

$$= 4x^2 - 3xy + (-4y^2 - 5y^2)$$

$$= 4x^2 - 3xy - 9y^2$$

$$\text{Hence, } 3x^2 + 5xy - 4y^2 + x^2 - 8xy - 5y^2 = 4x^2 - 3xy - 9y^2$$

Question 8(i)

Add :

$$5a + 3b, a - 2b, 3a + 5b$$

Answer

$$5a + 3b + a - 2b + 3a + 5b$$

Arrange the polynomial with like terms together and combine them.

$$= (5a + a + 3a) + (3b - 2b + 5b)$$

$$= 9a + 6b$$

Hence, addition of $5a + 3b, a - 2b, 3a + 5b$ is $9a + 6b$.

Question 8(ii)

Add :

$$8x - 3y + 7z, -4x + 5y - 4z, -x - y - 2z$$

Answer

$$8x - 3y + 7z + (-4x + 5y - 4z) + (-x - y - 2z)$$

$$= 8x - 3y + 7z - 4x + 5y - 4z - x - y - 2z$$

Arrange the polynomial with like terms together and combine them.

$$= (8x - 4x - x) + (-3y + 5y - y) + (7z - 4z - 2z)$$

$$= 3x + y + z$$

Hence, the addition of $8x - 3y + 7z, -4x + 5y - 4z, -x - y - 2z$ is $3x + y + z$.

Question 8(iii)

Add :

$$3b - 7c + 10, 5c - 2b - 15, 15 + 12c + b$$

Answer

$$3b - 7c + 10 + 5c - 2b - 15 + 15 + 12c + b$$

Arrange the polynomial with like terms together and combine them.

$$= (3b - 2b + b) + (-7c + 5c + 12c) + (10 - 15 + 15)$$

$$= 2b + 10c + 10$$

Hence, addition of $3b - 7c + 10, 5c - 2b - 15, 15 + 12c + b$ is $2b + 10c + 10$

Question 8(iv)

Add :

$$a - 3b + 3, 2a + 5 - 3c, 6c - 15 + 6b$$

Answer

$$a - 3b + 3 + 2a + 5 - 3c + 6c - 15 + 6b$$

Arrange the polynomial with like terms together and combine them.

$$= (a + 2a) + (-3b + 6b) + (3 + 5 - 15) + (-3c + 6c)$$

$$= 3a + 3b - 7 + 3c$$

Hence, the addition of $a - 3b + 3, 2a + 5 - 3c, 6c - 15 + 6b$ is $3a + 3b + 3c - 7$

Question 8(v)

Add :

$$13ab - 9cd - xy, 5xy, 15cd - 7ab, 6xy - 3cd$$

Answer

$$13ab - 9cd - xy + 5xy + 15cd - 7ab + 6xy - 3cd$$

Arrange the polynomial with like terms together and combine them.

$$= (13ab - 7ab) + (-9cd + 15cd - 3cd) + (-xy + 5xy + 6xy)$$

$$= 6ab + 3cd + 10xy$$

Hence, the addition of $13ab - 9cd - xy, 5xy, 15cd - 7ab, 6xy - 3cd$ is $6ab + 3cd + 10xy$.

Question 8(vi)

Add :

$$x^3 - x^2y + 5xy^2 + y^3, -x^3 - 9xy^2 + y^3, 3x^2y + 9xy^2$$

Answer

$$x^3 - x^2y + 5xy^2 + y^3 + (-x^3 - 9xy^2 + y^3) + 3x^2y + 9xy^2$$

Arrange the polynomial with like terms together and combine them.

$$= (x^3 - x^3) + (-x^2y + 3x^2y) + (5xy^2 - 9xy^2 + 9xy^2) + (y^3 + y^3)$$

$$= 0 + 2x^2y + 5xy^2 + 2y^3$$

Hence, the addition of $x^3 - x^2y + 5xy^2 + y^3$, $-x^3 - 9xy^2 + y^3$, $3x^2y + 9xy^2$ is $2x^2y + 5xy^2 + 2y^3$.

Question 9

Find the total savings of a boy who saves ₹ $(4x - 6y)$, ₹ $(6x + 2y)$, ₹ $(4y - x)$ and ₹ $(y - 2x)$ in four consecutive weeks.

Answer

The total savings = ₹ $(4x - 6y) + ₹ (6x + 2y) + ₹ (4y - x) + ₹ (y - 2x)$

$$= 4x - 6y + 6x + 2y + 4y - x + y - 2x$$

Arrange the polynomial with like terms together and combine them.

$$= (4x + 6x - x - 2x) + (-6y + 2y + 4y + y)$$

$$= 7x + y$$

Hence, the total saving is ₹ $(7x + y)$.

Question 10(i)

Subtract:

$$4xy^2 \text{ from } 3xy^2$$

Answer

$$3xy^2 - 4xy^2$$

$$= -xy^2$$

$$\text{Hence, } 3xy^2 - 4xy^2 = -1xy^2$$

Question 10(ii)

Subtract:

$$-2x^2y + 3xy^2 \text{ from } 8x^2y$$

Answer

$$8x^2y - (-2x^2y + 3xy^2)$$

$$= 8x^2y + 2x^2y - 3xy^2$$

Arrange the polynomial with like terms together and combine them.

$$= (8x^2y + 2x^2y) - 3xy^2$$

$$= 10x^2y - 3xy^2$$

$$\text{Hence, } 8x^2y - (-2x^2y + 3xy^2) = 10x^2y - 3xy^2$$

Question 10(iii)

Subtract:

$$3a - 5b + c + 2d \text{ from } 7a - 3b + c - 2d$$

Answer

$$7a - 3b + c - 2d - (3a - 5b + c + 2d)$$

$$= 7a - 3b + c - 2d - 3a + 5b - c - 2d$$

Arrange the polynomial with like terms together and combine them.

$$= (7a - 3a) + (-3b + 5b) + (c - c) + (-2d - 2d)$$

$$= 4a + 2b - 4d$$

$$\text{Hence, } 7a - 3b + c - 2d - (3a - 5b + c + 2d) = 4a + 2b - 4d.$$

Question 10(iv)

Subtract:

$$x^3 - 4x - 1 \text{ from } 3x^3 - x^2 + 6$$

Answer

$$3x^3 - x^2 + 6 - (x^3 - 4x - 1)$$

$$= 3x^3 - x^2 + 6 - x^3 + 4x + 1$$

Arrange the polynomial with like terms together and combine them.

$$= (3x^3 - x^3) - x^2 + 4x + (6 + 1)$$

$$= 2x^3 - x^2 + 4x + 7$$

$$\text{Hence, } 3x^3 - x^2 + 6 - (x^3 - 4x - 1) = 2x^3 - x^2 + 4x + 7$$

Question 10(v)

Subtract:

$$6a + 3 \text{ from } a^3 - 3a^2 + 4a + 1$$

Answer

$$a^3 - 3a^2 + 4a + 1 - (6a + 3)$$

$$= a^3 - 3a^2 + 4a + 1 - 6a - 3$$

Arrange the polynomial with like terms together and combine them.

$$= a^3 - 3a^2 + (4a - 6a) + (1 - 3)$$

$$= a^3 - 3a^2 - 2a - 2$$

$$\text{Hence, } a^3 - 3a^2 + 4a + 1 - (6a + 3) = a^3 - 3a^2 - 2a - 2.$$

Question 10(vi)

Subtract:

$$cab - 4cad - cbd \text{ from } 3abc + 5bcd - cda$$

Answer

$$3abc + 5bcd - cda - (cab - 4cad - cbd)$$

$$= 3abc + 5bcd - cda - cab + 4cad + cbd$$

Arrange the polynomial with like terms together and combine them.

$$= (3abc - abc) + (5bcd + bcd) + (-cda + 4cda)$$

$$= 2abc + 6bcd + 3cda$$

$$\text{Hence, } 3abc + 5bcd - cda - (cab - 4cad - cbd) = 2abc + 6bcd + 3cda.$$

Question 10(vii)

Subtract:

$$a^2 + ab + b^2 \text{ from } 4a^2 - 3ab + 2b^2$$

Answer

$$4a^2 - 3ab + 2b^2 - (a^2 + ab + b^2)$$

$$= 4a^2 - 3ab + 2b^2 - a^2 - ab - b^2$$

Arrange the polynomial with like terms together and combine them.

$$= (4a^2 - a^2) + (-3ab - ab) + (2b^2 - b^2)$$

$$= 3a^2 - 4ab + b^2$$

$$\text{Hence, } 4a^2 - 3ab + 2b^2 - (a^2 + ab + b^2) = 3a^2 - 4ab + b^2.$$

Question 11(i)

Take away $-3x^3 + 4x^2 - 5x + 6$ from $3x^3 - 4x^2 + 5x - 6$.

Answer

$$3x^3 - 4x^2 + 5x - 6 - (-3x^3 + 4x^2 - 5x + 6)$$

$$= 3x^3 - 4x^2 + 5x - 6 + 3x^3 - 4x^2 + 5x - 6$$

Arrange the polynomial with like terms together and combine them.

$$= (3x^3 + 3x^3) + (-4x^2 - 4x^2) + (5x + 5x) + (-6 - 6)$$

$$= 6x^3 - 8x^2 + 10x - 12$$

Hence, $3x^3 - 4x^2 + 5x - 6 - (-3x^3 + 4x^2 - 5x + 6) = 6x^3 - 8x^2 + 10x - 12$.

Question 11(ii)

Take $m^2 + m + 4$ from $-m^2 + 3m + 6$ and the result from $m^2 + m + 1$.

Answer

Difference between $m^2 + m + 4$ and $-m^2 + 3m + 6$

$$-m^2 + 3m + 6 - (m^2 + m + 4)$$

$$= -m^2 + 3m + 6 - m^2 - m - 4$$

Arrange the polynomial with like terms together and combine them.

$$= (-m^2 - m^2) + (3m - m) + (6 - 4)$$

$$= -2m^2 + 2m + 2$$

And, the difference between $-2m^2 + 2m + 2$ and $m^2 + m + 1$

$$(m^2 + m + 1) - (-2m^2 + 2m + 2)$$

$$= m^2 + m + 1 + 2m^2 - 2m - 2$$

Arrange the polynomial with like terms together and combine them.

$$= (m^2 + 2m^2) + (m - 2m) + (1 - 2)$$

$$= 3m^2 - m - 1$$

Hence, the result is $3m^2 - m - 1$.

Question 12

Subtract the sum of $5y^2 + y - 3$ and $y^2 - 3y + 7$ from $6y^2 + y - 2$.

Answer

The sum of $5y^2 + y - 3$ and $y^2 - 3y + 7$

$$5y^2 + y - 3 + (y^2 - 3y + 7)$$

$$= 5y^2 + y - 3 + y^2 - 3y + 7$$

Arrange the polynomial with like terms together and combine them.

$$= (5y^2 + y^2) + (y - 3y) + (-3 + 7)$$

$$= 6y^2 - 2y + 4$$

The difference of $6y^2 - 2y + 4$ from $6y^2 + y - 2$

$$= 6y^2 + y - 2 - (6y^2 - 2y + 4)$$

$$= 6y^2 + y - 2 - 6y^2 + 2y - 4$$

$$= (6y^2 - 6y^2) + (y + 2y) + (-2 - 4)$$

$$= 0 + 3y - 6$$

Hence, the result is $3y - 6$.

Question 13

What must be added to $x^4 - x^3 + x^2 + x + 3$ to obtain $x^4 + x^2 - 1$?

Answer

Let the number Z must be added to $x^4 - x^3 + x^2 + x + 3$ to obtain $x^4 + x^2 - 1$

$$\Rightarrow Z + (x^4 - x^3 + x^2 + x + 3) = (x^4 + x^2 - 1)$$

$$\Rightarrow Z = (x^4 + x^2 - 1) - (x^4 - x^3 + x^2 + x + 3)$$

$$\Rightarrow Z = x^4 + x^2 - 1 - x^4 + x^3 - x^2 - x - 3$$

Arrange the polynomial with like terms together and combine them.

$$\Rightarrow Z = (x^4 - x^4) + x^3 + (x^2 - x^2) - x + (-1 - 3)$$

$$\Rightarrow Z = 0 + x^3 + 0 - x - 4$$

Hence, the number is $x^3 - x - 4$.

Question 14(i)

How much more than $2x^2 + 4xy + 2y^2$ is $5x^2 + 10xy - y^2$?

Answer

Let $5x^2 + 10xy - y^2$ be more than $2x^2 + 4xy + 2y^2$ by Z.

$$2x^2 + 4xy + 2y^2 + Z = 5x^2 + 10xy - y^2$$

$$\Rightarrow Z = 5x^2 + 10xy - y^2 - (2x^2 + 4xy + 2y^2)$$

$$\Rightarrow Z = 5x^2 + 10xy - y^2 - 2x^2 - 4xy - 2y^2$$

Arrange the polynomial with like terms together and combine them.

$$\Rightarrow Z = (5x^2 - 2x^2) + (10xy - 4xy) + (-y^2 - 2y^2)$$

$$\Rightarrow Z = 3x^2 + 6xy - 3y^2$$

Hence, the number is $3x^2 + 6xy - 3y^2$.

Question 14(ii)

How much less $2a^2 + 1$ is than $3a^2 - 6$?

Answer

Let $2a^2 + 1$ be less than $3a^2 - 6$ by Z .

$$3a^2 - 6 - Z = 2a^2 + 1$$

$$\Rightarrow Z = 3a^2 - 6 - (2a^2 + 1)$$

$$\Rightarrow Z = 3a^2 - 6 - 2a^2 - 1$$

Arrange the polynomial with like terms together and combine them.

$$\Rightarrow Z = (3a^2 - 2a^2) + (-6 - 1)$$

$$\Rightarrow Z = a^2 - 7$$

Hence, the number is $a^2 - 7$.

Question 15

If $x = 6a + 8b + 9c$; $y = 2b - 3a - 6c$ and $z = c - b + 3a$; find :

(i) $x + y + z$

(ii) $x - y + z$

(iii) $2x - y - 3z$

(iv) $3y - 2z - 5x$

Answer

(i) If $x = 6a + 8b + 9c$; $y = 2b - 3a - 6c$ and $z = c - b + 3a$, then

$$x + y + z = (6a + 8b + 9c) + (2b - 3a - 6c) + (c - b + 3a)$$

$$= 6a + 8b + 9c + 2b - 3a - 6c + c - b + 3a$$

Arrange the polynomial with like terms together and combine them.

$$= (6a - 3a + 3a) + (8b + 2b - b) + (9c - 6c + c)$$

$$= 6a + 9b + 4c$$

Hence, $x + y + z = 6a + 9b + 4c$

(ii) If $x = 6a + 8b + 9c$; $y = 2b - 3a - 6c$ and $z = c - b + 3a$, then

$$x - y + z = (6a + 8b + 9c) - (2b - 3a - 6c) + (c - b + 3a)$$

$$= 6a + 8b + 9c - 2b + 3a + 6c + c - b + 3a$$

Arrange the polynomial with like terms together and combine them.

$$= (6a + 3a + 3a) + (8b - 2b - b) + (9c + 6c + c)$$

$$= 12a + 5b + 16c$$

Hence, $x - y + z = 12a + 5b + 16c$

(iii) If $x = 6a + 8b + 9c$; $y = 2b - 3a - 6c$ and $z = c - b + 3a$, then

$$2x - y - 3z = 2(6a + 8b + 9c) - (2b - 3a - 6c) - 3(c - b + 3a)$$

$$= 12a + 16b + 18c - 2b + 3a + 6c - 3c + 3b - 9a$$

Arrange the polynomial with like terms together and combine them.

$$= (12a + 3a - 9a) + (16b - 2b + 3b) + (18c + 6c - 3c)$$

$$= 6a + 17b + 21c$$

Hence, $2x - y - 3z = 6a + 17b + 21c$

(iv) If $x = 6a + 8b + 9c$; $y = 2b - 3a - 6c$ and $z = c - b + 3a$, then

$$3y - 2z - 5x = 3(2b - 3a - 6c) - 2(c - b + 3a) - 5(6a + 8b + 9c)$$

$$= 6b - 9a - 18c - 2c + 2b - 6a - 30a - 40b - 45c$$

Arrange the polynomial with like terms together and combine them.

$$= (-9a - 6a - 30a) + (6b + 2b - 40b) + (-18c - 2c - 45c)$$

$$= -45a - 32b - 65c$$

Hence, $3y - 2z - 5x = -45a - 32b - 65c$

Exercise 11(B)

Question 1(i)

$(9x^4 - 8x^3 - 12x) \times (3x)$ is equal to:

1. $27x^5 - 24x^4 + 36x^2$

2. $27x^5 - 24x^4 - 36x^2$

3. $27x^5 + 24x^4 - 36x^2$

4. $27x^5 + 24x^4 + 36x^2$

Answer

$$(3x) \times (9x^4 - 8x^3 - 12x)$$

$$= 3x \times 9x^4 - 3x \times 8x^3 - 3x \times 12x$$

$$= 27x^{(4+1)} - 24x^{(3+1)} - 36x^{(1+1)}$$

$$= 27x^5 - 24x^4 - 36x^2$$

Hence, option 2 is the correct option.

Question 1(ii)

$(9x^4 - 12x^3 - 18x) \div (3x)$ is equal to:

1. $3x^3 + 4x^2 + 6$

2. $3x^3 + 4x^2 - 6$

3. $3x^3 - 4x^2 - 6$

4. $3x^4 - 4x^2 + 6$

Answer

$$(9x^4 - 12x^3 - 18x) \div (3x)$$

$$= (9x^4 \div 3x) - (12x^3 \div 3x) - (18x \div 3x)$$

$$= 3x^{(4-1)} - 4x^{(3-1)} - 6x^{(1-1)}$$

$$= 3x^3 - 4x^2 - 6$$

Hence, option 3 is the correct option.

Question 1(iii)

$\left(\frac{10}{3}xy^2z\right) \times \left(-\frac{9}{5}x^2z\right)$ is equal to:

1. $-6x^3y^2z^2$

2. $6x^3y^2z^2$

$$3. -\frac{2}{3}x^2y^2z^2$$

$$4. \frac{9}{5}xy^3z^2$$

Answer

$$\begin{aligned} & \left(\frac{10}{3}xy^2z\right) \times \left(-\frac{9}{5}x^2z\right) \\ &= \left(-\frac{10 \times 9}{3 \times 5}x^{1+2}y^2z^{1+1}\right) \\ &= \left(-\frac{90}{15}x^3y^2z^2\right) \\ &= (-6x^3y^2z^2) \end{aligned}$$

Hence, option 1 is the correct option.

Question 1(iv)

$(x^3 + y^2) \times 10x^2$ is equal to :

1. $10x^5 + 10xy$

2. $10x^5 + 10x^2y^2$

3. $10x^5 - 10x^2y^2$

4. $-10x^5 - 10x^2y^2$

Answer

$$(x^3 + y^2) \times 10x^2$$

$$= 10x^2 \times x^3 + 10x^2 \times y^2$$

$$= 10x^{(2+3)} + 10x^2y^2$$

$$= 10x^5 + 10x^2y^2$$

Hence, option 2 is the correct option.

Question 1(v)

$(a^3 - b^3) \div (a - b)$ is equal to :

1. $a^2 + b^2 + ab$

2. $a^2 + b^2 - ab$

3. $a^2 - b^2 + ab$

4. $a^2 - b^2 - ab$

Answer

$$(a^3 - b^3) \div (a - b)$$

$$[\because \text{As we know, } (x^3 - y^3) = (x - y)(x^2 + y^2 + xy)]$$

$$\therefore (a^3 - b^3) \div (a - b) = \frac{(a - b)(a^2 + b^2 + ab)}{(a - b)}$$

$$= \frac{\cancel{(a - b)}(a^2 + b^2 + ab)}{\cancel{(a - b)}}$$

$$= (a^2 + b^2 + ab)$$

Hence, option 1 is the correct option.

Question 2(i)

Multiply :

$$8ab^2 \text{ by } -4a^3b^4$$

Answer

$$8ab^2 \times (-4a^3b^4)$$

$$= 8 \times (-4) a^{(1+3)} b^{(2+4)}$$

$$= -32 a^4 b^6$$

$$\text{Hence, } 8ab^2 \times (-4a^3b^4) = -32a^4 b^6$$

Question 2(ii)

Multiply :

$$\frac{2}{3} ab \text{ by } -\frac{1}{4} a^2b$$

Answer

$$\frac{2}{3} ab \times \left(-\frac{1}{4} a^2b \right)$$

$$= -\frac{2 \times 1}{3 \times 4} a^{1+2} b^{1+1}$$

$$= -\frac{2}{12}a^3b^2$$

$$= -\frac{1}{6}a^3b^2$$

$$\text{Hence, } \frac{2}{3}ab \times \left(-\frac{1}{4}a^2b\right) = -\frac{1}{6}a^3b^2.$$

Question 2(iii)

Multiply :

$$-5cd^2 \text{ by } -5cd^2$$

Answer

$$(-5cd^2) \times (-5cd^2)$$

$$= (-5) \times (-5) c^{(1+1)}d^{(2+2)}$$

$$= 25c^2d^4$$

$$\text{Hence, } (-5cd^2) \times (-5cd^2) = 25c^2d^4$$

Question 2(iv)

Multiply :

$$4a \text{ and } 6a + 7$$

Answer

$$4a \times (6a + 7)$$

$$= 4a \times 6a + 4a \times 7$$

$$= 24a^{(1+1)} + 28a$$

$$= 24a^2 + 28a$$

$$\text{Hence, } 4a \times (6a + 7) = 24a^2 + 28a$$

Question 2(v)

Multiply :

$$-8x \text{ and } 4 - 2x - x^2$$

Answer

$$-8x \times (4 - 2x - x^2)$$

$$= (-8x) \times 4 - (-8x) \times 2x - (-8x) \times x^2$$

$$= -32x + 16x^{(1+1)} + 8x^{(1+2)}$$

$$= -32x + 16x^2 + 8x^3$$

$$\text{Hence, } (-8x) \times (4 - 2x - x^2) = -32x + 16x^2 + 8x^3$$

Question 2(vi)

Multiply :

$$2a^2 - 5a - 4 \text{ and } -3a$$

Answer

$$-3a \times (2a^2 - 5a - 4)$$

$$= -3a \times 2a^2 - (-3a) \times 5a - (-3a) \times 4$$

$$= -6a^{(2+1)} + 15a^{(1+1)} + 12a$$

$$= -6a^3 + 15a^2 + 12a$$

$$\text{Hence, } (2a^2 - 5a - 4) \times (-3a) = -6a^3 + 15a^2 + 12a$$

Question 2(vii)

Multiply :

$$x + 4 \text{ by } x - 5$$

Answer

$$(x + 4) \times (x - 5)$$

$$= x \times (x - 5) + 4 \times (x - 5)$$

$$= x \times x - x \times 5 + 4 \times x - 4 \times 5$$

$$= x^2 - 5x + 4x - 20$$

$$= x^2 - x - 20$$

$$\text{Hence, } (x + 4) \times (x - 5) = x^2 - x - 20$$

Question 2(viii)

Multiply :

$$5a - 1 \text{ by } 7a - 3$$

Answer

$$\begin{aligned}(5a - 1) \times (7a - 3) \\&= 5a \times (7a - 3) - 1 \times (7a - 3) \\&= 5a \times 7a - 5a \times 3 - 1 \times 7a - 1 \times (-3) \\&= 35a^{(1+1)} - 15a - 7a + 3 \\&= 35a^2 - 22a + 3\end{aligned}$$

$$\text{Hence, } (5a - 1) \times (7a - 3) = 35a^2 - 22a + 3$$

Question 2(ix)

Multiply :

$$12a + 5b \text{ by } 7a - b$$

Answer

$$\begin{aligned}(12a + 5b) \times (7a - b) \\&= 12a \times 7a - 12a \times b + 5b \times 7a - 5b \times b \\&= 84a^{(1+1)} - 12ab + 35ab - 5b^{(1+1)} \\&= 84a^2 + 23ab - 5b^2\end{aligned}$$

$$\text{Hence, } (12a + 5b) \times (7a - b) = 84a^2 + 23ab - 5b^2$$

Question 2(x)

Multiply :

$$x^2 + x + 1 \text{ by } 1 - x$$

Answer

$$\begin{aligned}(x^2 + x + 1) \times (1 - x) \\&= 1 \times (x^2 + x + 1) - x \times (x^2 + x + 1) \\&= (1 \times x^2 + 1 \times x + 1 \times 1) - (x \times x^2 + x \times x + x \times 1) \\&= x^2 + x + 1 - (x^3 + x^2 + x) \\&= x^2 + x + 1 - x^3 - x^2 - x \\&= (x^2 - x^2) + (x - x) + 1 - x^3 \\&= 1 - x^3\end{aligned}$$

$$\text{Hence, } (x^2 + x + 1) \times (1 - x) = 1 - x^3$$

Question 2(xi)

Multiply :

$$2m^2 - 3m - 1 \text{ and } 4m^2 - m - 1$$

Answer

$$(2m^2 - 3m - 1) \times (4m^2 - m - 1)$$

$$= 2m^2 \times 4m^2 - 2m^2 \times m - 2m^2 \times 1 - 3m \times 4m^2 - 3m \times (-m) - 3m \times (-1) - 1 \times 4m^2 - 1 \times (-m) - 1 \times (-1)$$

$$= 8m^{(2+2)} - 2m^{(2+1)} - 2m^2 - 12m^{(2+1)} + 3m^{(1+1)} + 3m - 4m^2 + m + 1$$

$$= 8m^4 - 2m^3 - 2m^2 - 12m^3 + 3m^2 + 3m - 4m^2 + m + 1$$

$$= 8m^4 - 2m^3 - 12m^3 - 2m^2 + 3m^2 - 4m^2 + 3m + m + 1$$

$$= 8m^4 - 14m^3 - 3m^2 + 4m + 1$$

$$\text{Hence, } (2m^2 - 3m - 1) \times (4m^2 - m - 1) = 8m^4 - 14m^3 - 3m^2 + 4m + 1$$

Question 2(xii)

Multiply :

$$a^2, ab \text{ and } b^2$$

Answer

$$(a^2 \times ab) \times b^2$$

$$= a^{(2+1)}b \times b^2$$

$$= a^3b \times b^2$$

$$= a^3b^{(2+1)}$$

$$= a^3b^3$$

$$\text{Hence, } a^2 \times ab \times b^2 = a^3b^3$$

Question 2(xiii)

Multiply :

$$abx, -3a^2x \text{ and } 7b^2x^3$$

Answer

$$\begin{aligned} & abx \times (-3a^2x) \times 7b^2x^3 \\ &= (-3 \times 7) a^{(1+2)} b^{(1+2)} x^{(1+1+3)} \\ &= -21 a^3 b^3 x^5 \end{aligned}$$

$$\text{Hence, } (abx) \times (-3a^2x) \times (7b^2x^3) = -21a^3b^3x^5$$

Question 2(xiv)

Multiply :

$$-3bx, -5xy \text{ and } -7b^3y^2$$

Answer

$$\begin{aligned} & -3bx \times (-5xy) \times (-7b^3y^2) \\ &= -3 \times (-5) \times (-7) b^{(1+3)} x^{(1+1)} y^{(1+2)} \\ &= -105 b^4 x^2 y^3 \end{aligned}$$

$$\text{Hence, } (-3bx) \times (-5xy) \times (-7b^3y^2) = -105b^4x^2y^3$$

Question 2(xv)

Multiply :

$$-\frac{3}{2}x^5y^3 \text{ and } \frac{4}{9}a^2x^3y$$

Answer

$$\begin{aligned} & -\frac{3}{2}x^5y^3 \times \frac{4}{9}a^2x^3y \\ &= -\frac{3 \times 4}{2 \times 9}a^2x^{5+3}y^{3+1} \\ &= -\frac{12}{18}a^2x^8y^4 \\ &= -\frac{2}{3}a^2x^8y^4 \end{aligned}$$

$$\text{Hence, } -\frac{3}{2}x^5y^3 \times \frac{4}{9}a^2x^3y = -\frac{2}{3}a^2x^8y^4$$

Question 2(xvi)

Multiply :

$$-\frac{2}{3}a^7b^2 \text{ and } -\frac{9}{4}ab^5$$

Answer

$$\begin{aligned} & -\frac{2}{3}a^7b^2 \times \left(-\frac{9}{4}ab^5\right) \\ &= \frac{2 \times 9}{3 \times 4}a^{7+1}b^{2+5} \end{aligned}$$

$$= \frac{18}{12}a^8b^7$$

$$= \frac{3}{2}a^8b^7$$

$$\text{Hence, } -\frac{2}{3}a^7b^2 \times -\frac{9}{4}ab^5 = \frac{3}{2}a^8b^7$$

Question 2(xvii)

Multiply :

$$2a^3 - 3a^2b \text{ and } -\frac{1}{2}ab^2$$

Answer

$$(2a^3 - 3a^2b) \times \left(-\frac{1}{2}ab^2\right)$$

$$= \left[2a^3 \times \left(-\frac{1}{2}ab^2\right) - 3a^2b \times \left(-\frac{1}{2}ab^2\right)\right]$$

$$= -\frac{1 \times 2}{2}a^{3+1}b^2 + \frac{1 \times 3}{2}a^{2+1}b^{1+2}$$

$$= -\frac{2}{2}a^4b^2 + \frac{3}{2}a^3b^3$$

$$= -a^4b^2 + \frac{3}{2}a^3b^3$$

$$\text{Hence, } 2a^3 - 3a^2b \times -\frac{1}{2}ab^2 = -a^4b^2 + \frac{3}{2}a^3b^3$$

Question 2(xviii)

Multiply :

$$2x + \frac{1}{2}y \text{ and } 2x - \frac{1}{2}y$$

Answer

$$\begin{aligned} & \left(2x + \frac{1}{2}y\right) \times \left(2x - \frac{1}{2}y\right) \\ &= 2x \times \left(2x - \frac{1}{2}y\right) + \frac{1}{2}y \times \left(2x - \frac{1}{2}y\right) \\ &= \left(2x \times 2x - 2x \times \frac{1}{2}y\right) + \left(\frac{1}{2}y \times 2x - \frac{1}{2}y \times \frac{1}{2}y\right) \\ &= \left(4x^{1+1} - \frac{2xy}{2}\right) + \left(\frac{2xy}{2} - \frac{1 \times 1}{2 \times 2}y^{1+1}\right) \\ &= (4x^2 - xy) + \left(xy - \frac{1}{4}y^2\right) \\ &= 4x^2 - xy + xy - \frac{1}{4}y^2 \\ &= 4x^2 - \cancel{xy} + \cancel{xy} - \frac{1}{4}y^2 \\ &= 4x^2 - \frac{1}{4}y^2 \end{aligned}$$

$$\text{Hence, } 2x + \frac{1}{2}y \times 2x - \frac{1}{2}y = 4x^2 - \frac{1}{4}y^2$$

Question 3(i)

Multiply :

$$5x^2 - 8xy + 6y^2 - 3 \text{ by } -3xy$$

Answer

$$(5x^2 - 8xy + 6y^2 - 3) \times (-3xy)$$

$$= 5x^2 \times (-3xy) - 8xy \times (-3xy) + 6y^2 \times (-3xy) - 3 \times (-3xy)$$

$$= -15x^{(2+1)}y + 24x^{(1+1)}y^{(1+1)} - 18xy^{(2+1)} + 9xy$$

$$= -15x^3y + 24x^2y^2 - 18xy^3 + 9xy$$

$$\text{Hence, } (5x^2 - 8xy + 6y^2 - 3) \times (-3xy) = -15x^3y + 24x^2y^2 - 18xy^3 + 9xy$$

Question 3(ii)

Multiply :

$$3 - \frac{2}{3}xy + \frac{5}{7}xy^2 - \frac{16}{21}x^2y \text{ by } -21x^2y^2$$

Answer

$$\left(3 - \frac{2}{3}xy + \frac{5}{7}xy^2 - \frac{16}{21}x^2y\right) \times (-21x^2y^2)$$

$$= \left(3 \times (-21x^2y^2) - \frac{2}{3}xy \times (-21x^2y^2) + \frac{5}{7}xy^2 \times (-21x^2y^2) - \frac{16}{21}x^2y \times (-21x^2y^2)\right)$$

$$= \left(-63x^2y^2 - \frac{2 \times (-21)}{3}x^{1+2}y^{1+2} + \frac{5 \times (-21)}{7}x^{1+2}y^{2+2} - \frac{16 \times (-21)}{21}x^{2+2}y^{1+2}\right)$$

$$= -63x^2y^2 - \frac{-42}{3}x^3y^3 + \frac{-105}{7}x^3y^4 + \frac{16 \times \cancel{21}}{\cancel{21}}x^4y^3$$

$$= -63x^2y^2 + 14x^3y^3 - 15x^3y^4 + 16x^4y^3$$

Hence, $(3 - \frac{2}{3}xy + \frac{5}{7}xy^2 - \frac{16}{21}x^2y) \times (-21x^2y^2) = -63x^2y^2 + 14x^3y^3 - 15x^3y^4 + 16x^4y^3$

Question 3(iii)

Multiply :

$$6x^3 - 5x + 10 \text{ by } 4 - 3x^2$$

Answer

$$(6x^3 - 5x + 10) \times (4 - 3x^2)$$

$$= 4 \times (6x^3 - 5x + 10) - 3x^2 \times (6x^3 - 5x + 10)$$

$$= 24x^3 - 20x + 40 - 18x^{(3+2)} + 15x^{(1+2)} - 30x^2$$

$$= 24x^3 - 20x + 40 - 18x^5 + 15x^3 - 30x^2$$

$$= -18x^5 + 24x^3 + 15x^3 - 30x^2 - 20x + 40$$

$$= -18x^5 + 39x^3 - 30x^2 - 20x + 40$$

Hence, $(6x^3 - 5x + 10) \times (4 - 3x^2) = -18x^5 + 39x^3 - 30x^2 - 20x + 40$

Question 3(iv)

Multiply :

$$2y - 4y^3 + 6y^5 \text{ by } y^2 + y - 3$$

Answer

$$(2y - 4y^3 + 6y^5) \times (y^2 + y - 3)$$

$$= 2y \times (y^2 + y - 3) - 4y^3 \times (y^2 + y - 3) + 6y^5 \times (y^2 + y - 3)$$

$$= 2y^{(1+2)} + 2y^{(1+1)} - 6y - 4y^{(3+2)} - 4y^{(3+1)} + 12y^3 + 6y^{(5+2)} + 6y^{(5+1)} - 18y^5$$

$$= 2y^3 + 2y^2 - 6y - 4y^5 - 4y^4 + 12y^3 + 6y^7 + 6y^6 - 18y^5$$

$$= 6y^7 + 6y^6 - 18y^5 - 4y^5 - 4y^4 + 12y^3 + 2y^3 + 2y^2 - 6y$$

$$= 6y^7 + 6y^6 - 22y^5 - 4y^4 + 14y^3 + 2y^2 - 6y$$

$$\text{Hence, } (2y - 4y^3 + 6y^5) \times (y^2 + y - 3) = 6y^7 + 6y^6 - 22y^5 - 4y^4 + 14y^3 + 2y^2 - 6y$$

Question 3(v)

Multiply :

$$5p^2 + 25pq + 4q^2 \text{ by } 2p^2 - 2pq + 3q^2$$

Answer

$$(5p^2 + 25pq + 4q^2) \times (2p^2 - 2pq + 3q^2)$$

$$= 5p^2 \times (2p^2 - 2pq + 3q^2) + 25pq \times (2p^2 - 2pq + 3q^2) + 4q^2 \times (2p^2 - 2pq + 3q^2)$$

$$= 10p^{(2+2)} - 10p^{(2+1)}q + 15p^2q^2 + 50p^{(1+2)}q - 50p^{(1+1)}q^{(1+1)} + 75pq^{(1+2)} + 8p^2q^2$$

$$- 8pq^{(2+1)} + 12q^{(2+2)}$$

$$= 10p^4 - 10p^3q + 15p^2q^2 + 50p^3q - 50p^2q^2 + 75pq^3 + 8p^2q^2 - 8pq^3 + 12q^4$$

$$= 10p^4 - 10p^3q + 50p^3q + 15p^2q^2 - 50p^2q^2 + 8p^2q^2 + 75pq^3 - 8pq^3 + 12q^4$$

$$= 10p^4 + 40p^3q - 27p^2q^2 + 67pq^3 + 12q^4$$

Hence, $(5p^2 + 25pq + 4q^2) \times (2p^2 - 2pq + 3q^2) = 10p^4 + 40p^3q - 27p^2q^2 + 67pq^3 + 12q^4$

Question 4(i)

Simplify:

$$(7x - 8)(3x + 2)$$

Answer

$$(7x - 8)(3x + 2)$$

$$= 7x(3x + 2) - 8(3x + 2)$$

$$= 21x^{(1+1)} + 14x - 24x - 16$$

$$= 21x^2 + 14x - 24x - 16$$

$$= 21x^2 - 10x - 16$$

Hence, $(7x - 8)(3x + 2) = 21x^2 - 10x - 16$

Question 4(ii)

Simplify:

$$(px - q)(px + q)$$

Answer

$$(px - q)(px + q)$$

$$= px(px + q) - q(px + q)$$

$$= p^{(1+1)}x^{(1+1)} + pqx - pqx - q^{(1+1)}$$

$$= p^2x^2 + pqx - pqx - q^2$$

$$= p^2x^2 - q^2$$

$$\text{Hence, } (px - q)(px + q) = p^2x^2 - q^2$$

Question 4(iii)

Simplify:

$$(5a + 5b - c)(2b - 3c)$$

Answer

$$(5a + 5b - c)(2b - 3c)$$

$$= 5a(2b - 3c) + 5b(2b - 3c) - c(2b - 3c)$$

$$= 10ab - 15ac + 10b^{(1+1)} - 15bc - 2bc + 3c^{(1+1)}$$

$$= 10ab - 15ac + 10b^2 - 17bc + 3c^2$$

$$\text{Hence, } (5a + 5b - c)(2b - 3c) = 10ab - 15ac + 10b^2 - 17bc + 3c^2$$

Question 4(iv)

Simplify:

$$(4x - 5y)(5x - 4y)$$

Answer

$$(4x - 5y)(5x - 4y)$$

$$= 4x(5x - 4y) - 5y(5x - 4y)$$

$$= 20x^{(1+1)} - 16xy - 25xy + 20y^{(1+1)}$$

$$= 20x^2 - 41xy + 20y^2$$

$$\text{Hence, } (4x - 5y)(5x - 4y) = 20x^2 - 41xy + 20y^2$$

Question 4(v)

Simplify:

$$(3y + 4z)(3y - 4z) + (2y + 7z)(y + z)$$

Answer

$$(3y + 4z)(3y - 4z) + (2y + 7z)(y + z)$$

$$= [3y(3y - 4z) + 4z(3y - 4z)] + [2y(y + z) + 7z(y + z)]$$

$$= [9y^2 - 12yz + 12yz - 16z^2] + [2y^2 + 2yz + 7zy + 7z^2]$$

$$= 9y^2 - 16z^2 + 2y^2 + 9zy + 7z^2$$

$$= 9y^2 + 2y^2 + 9zy - 16z^2 + 7z^2$$

$$= 11y^2 + 9yz - 9z^2$$

$$\text{Hence, } (3y + 4z)(3y - 4z) + (2y + 7z)(y + z) = 11y^2 + 9yz - 9z^2$$

Question 5

The adjacent sides of a rectangle are $x^2 - 4xy + 7y^2$ and $x^3 - 5xy^2$. Find its area.

Answer

Given:

Dimensions of the rectangle = $x^2 - 4xy + 7y^2$ and $x^3 - 5xy^2$

As we know, area of rectangle = l x b

So, putting the value

$$\begin{aligned} & (x^2 - 4xy + 7y^2) \times (x^3 - 5xy^2) \\ &= x^2 \times (x^3 - 5xy^2) - 4xy \times (x^3 - 5xy^2) + 7y^2 \times (x^3 - 5xy^2) \\ &= x^{(2+3)} - 5x^{(2+1)}y^2 - 4x^{(1+3)}y + 20x^{(1+1)}y^{(1+2)} + 7x^3y^2 - 35xy^{(2+2)} \\ &= x^5 - 5x^3y^2 + 7x^3y^2 - 4x^4y + 20x^2y^3 - 35xy^4 \\ &= x^5 + 2x^3y^2 - 4x^4y + 20x^2y^3 - 35xy^4 \end{aligned}$$

$$\text{Hence, the area of the rectangle} = x^5 + 2x^3y^2 - 4x^4y + 20x^2y^3 - 35xy^4$$

Question 6

The base and the altitude of a triangle are $(3x - 4y)$ and $(6x + 5y)$ respectively.
Find its area.

Answer

Given:

Base of a triangle = $(3x - 4y)$

Altitude of a triangle = $(6x + 5y)$

As we know that the area of triangle = $\frac{1}{2} \times \text{base} \times \text{altitude}$

$$\frac{1}{2} \times (3x - 4y) \times (6x + 5y)$$

$$= \frac{1}{2} \times (3x(6x + 5y) - 4y(6x + 5y))$$

$$= \frac{1}{2} \times (18x^2 + 15xy - 24xy - 20y^2)$$

$$= \frac{1}{2} \times (18x^2 - 9xy - 20y^2)$$

$$\text{Hence, the area of triangle} = \frac{1}{2} (18x^2 - 9xy - 20y^2) \text{ sq. unit}$$

Question 7

Multiply $-4xy^3$ and $6x^2y$ and verify your result for $x = 2$ and $y = 1$.

Answer

$$(-4xy^3) \times (6x^2y)$$

$$= - 24x^{(1+2)}y^{(3+1)}$$

$$= - 24x^3y^4$$

$$\text{Hence, } (- 4xy^3) \times (6x^2y) = - 24x^3y^4$$

Putting $x = 2$ and $y = 1$

Taking LHS:

$$(- 4xy^3) \times (6x^2y)$$

$$= (- 4 \times 2 \times 1^3) \times (6 \times 2^2 \times 1)$$

$$= (- 4 \times 2 \times 1) \times (6 \times 4 \times 1)$$

$$= (- 8) \times (24)$$

$$= - 192$$

Taking RHS:

$$- 24x^3y^4$$

$$= - 24 \times 2^3 \times 1^4$$

$$= - 24 \times 8 \times 1$$

$$= - 192$$

Hence, LHS = RHS

$$\text{So, } (- 4xy^3) \times (6x^2y) = - 24x^3y^4$$

Question 8

Find the value of $(3x^3) \times (-5xy^2) \times (2x^2yz^3)$ for $x = 1$, $y = 2$ and $z = 3$.

Answer

$$(3x^3) \times (-5xy^2) \times (2x^2yz^3)$$

$$= (-15x^{(3+1)}y^2) \times (2x^2yz^3)$$

$$= (-15x^4y^2) \times (2x^2yz^3)$$

$$= -30x^{(4+2)}y^{(2+1)}z^3$$

$$= -30x^6y^3z^3$$

Putting the value $x = 1$, $y = 2$ and $z = 3$, we get

$$-30 \times 1^6 \times 2^3 \times 3^3$$

$$= -30 \times 1 \times 8 \times 27$$

$$= -6480$$

Hence, the value of $(3x^3) \times (-5xy^2) \times (2x^2yz^3)$ for $x = 1$, $y = 2$ and $z = 3$ is - 6480

Question 9

Evaluate $(3x^4y^2) (2x^2y^3)$ for $x = 1$ and $y = 2$.

Answer

$$(3x^4y^2) \times (2x^2y^3)$$

$$= 6x^{(4+2)}y^{(2+3)}$$

$$= 6x^6y^5$$

Putting $x = 1$ and $y = 2$, we get

$$6 \times 1^6 \times 2^5$$

$$= 6 \times 1 \times 32$$

$$= 192$$

Hence, $(3x^4y^2) (2x^2y^3)$ for $x = 1$ and $y = 2$ is 192

Question 10

Evaluate $(x^5) \times (3x^2) \times (-2x)$ for $x = 1$.

Answer

Given:

$$(x^5) \times (3x^2) \times (-2x)$$

$$= 3x^{(5+2)} \times (-2x)$$

$$= 3x^7 \times (-2x)$$

$$= -6x^{(7+1)}$$

$$= -6x^8$$

For $x = 1$

$$- 6 \times 1^8$$

$$= - 6 \times 1$$

$$= - 6$$

Hence, the value of $(x^5) \times (3x^2) \times (-2x)$ for $x = 1$ is -6

Question 11(i)

Divide:

$$- 70a^3 \text{ by } 14a^2$$

Answer

$$- 70a^3 \div 14a^2$$

$$= - \frac{70}{14} a^{(3-2)}$$

$$= - 5 a^1$$

Hence, $- 70a^3 \div 14a^2 = - 5 a$

Question 11(ii)

Divide:

$$24x^3y^3 \text{ by } - 8y^2$$

Answer

$$(24x^3y^3) \div (-8y^2)$$

$$= -\frac{24}{8} x^3 y^{(3-2)}$$

$$= -3 x^3 y^1$$

$$\text{Hence, } (24x^3y^3) \div (-8y^2) = -3 x^3 y$$

Question 11(iii)

Divide:

$$15a^4b \text{ by } -5a^3b$$

Answer

$$(15a^4b) \div (-5a^3b)$$

$$= -\frac{15}{5} a^{(4-3)} b^{(1-1)}$$

$$= -3a^1 b^0$$

$$\text{Hence, } (15a^4b) \div (-5a^3b) = -3a$$

Question 11(iv)

Divide:

$$-24x^4d^3 \text{ by } -2x^2d^5$$

Answer

$$(-24x^4d^3) \div (-2x^2d^5)$$

$$= \frac{24}{2} x^{(4-2)} d^{(3-5)}$$

$$= 12 x^2 d^{(-2)}$$

$$= \frac{12x^2}{d^2}$$

$$\text{Hence, } (-24x^4d^3) \div (-2x^2d^5) = \frac{12x^2}{d^2}$$

Question 11(v)

Divide:

$$63a^4b^5c^6 \text{ by } -9a^2b^4c^3$$

Answer

$$(63a^4b^5c^6) \div (-9a^2b^4c^3)$$

$$= -\frac{63}{9} a^{(4-2)} b^{(5-4)} c^{(6-3)}$$

$$= -7a^2b^1c^3$$

$$\text{Hence, } (63a^4b^5c^6) \div (-9a^2b^4c^3) = -7a^2bc^3$$

Question 11(vi)

Divide:

$$8x - 10y + 6c \text{ by } 2$$

Answer

$$\begin{aligned}(8x - 10y + 6c) \div 2 \\&= (8x \div 2) - (10y \div 2) + (6c \div 2) \\&= 4x - 5y + 3c\end{aligned}$$

$$\text{Hence, } (8x - 10y + 6c) \div 2 = 4x - 5y + 3c$$

Question 11(vii)

Divide:

$$15a^3b^4 - 10a^4b^3 - 25a^3b^6 \text{ by } -5a^3b^2$$

Answer

$$\begin{aligned}(15a^3b^4 - 10a^4b^3 - 25a^3b^6) \div (-5a^3b^2) \\&= (15a^3b^4 \div (-5a^3b^2)) - (10a^4b^3 \div (-5a^3b^2)) - (25a^3b^6 \div (-5a^3b^2)) \\&= \left(-\frac{15}{5} a^{(3-3)}b^{(4-2)}\right) - \left(-\frac{10}{5} a^{(4-3)}b^{(3-2)}\right) - \left(-\frac{25}{5} a^{(3-3)}b^{(6-2)}\right) \\&= -3 a^0b^2 + 2 a^1b^1 + 5 a^0b^4 \\&= -3b^2 + 2ab + 5b^4\end{aligned}$$

$$\text{Hence, } (15a^3b^4 - 10a^4b^3 - 25a^3b^6) \div (-5a^3b^2) = -3b^2 + 2ab + 5b^4$$

Question 11(viii)

Divide:

$$-14x^6y^3 - 21x^4y^5 + 7x^5y^4 \text{ by } 7x^2y^2$$

Answer

$$\begin{aligned}
 & (-14x^6y^3 - 21x^4y^5 + 7x^5y^4) \div 7x^2y^2 \\
 &= (-14x^6y^3 \div 7x^2y^2) - (21x^4y^5 \div 7x^2y^2) + (7x^5y^4 \div 7x^2y^2) \\
 &= \left(-\frac{14}{7} x^{(6-2)}y^{(3-2)} \right) - \left(\frac{21}{7} x^{(4-2)}y^{(5-2)} \right) - \left(-\frac{7}{7} x^{(5-2)}y^{(4-2)} \right) \\
 &= -2x^4y^1 - 3x^2y^3 + 1x^3y^2
 \end{aligned}$$

$$\text{Hence, } (-14x^6y^3 - 21x^4y^5 + 7x^5y^4) \div 7x^2y^2 = -2x^4y - 3x^2y^3 + x^3y^2$$

Question 11(ix)

Divide:

$$a^2 + 7a + 12 \text{ by } a + 4$$

Answer

$$\begin{aligned}
 & (a^2 + 7a + 12) \div (a + 4) \\
 &= (a^2 + 3a + 4a + 12) \div (a + 4) \\
 &= [a(a + 3) + 4(a + 3)] \div (a + 4) \\
 &= [(a + 3)(a + 4)] \div (a + 4) \\
 &= \frac{(a + 3)\cancel{(a + 4)}}{\cancel{(a + 4)}} \\
 &= (a + 3)
 \end{aligned}$$

Hence, $(a^2 + 7a + 12) \div (a + 4) = a + 3$

Question 11(x)

Divide:

$x^2 + 3x - 54$ by $x - 6$

Answer

$$(x^2 + 3x - 54) \div (x - 6)$$

$$= (x^2 + 9x - 6x - 54) \div (x - 6)$$

$$= [x(x + 9) - 6(x + 9)] \div (x - 6)$$

$$= [(x + 9)(x - 6)] \div (x - 6)$$

$$= \frac{(x + 9)\cancel{(x - 6)}}{\cancel{(x - 6)}}$$

$$= (x + 9)$$

Hence, $(x^2 + 3x - 54) \div (x - 6) = x + 9$

Question 11(xi)

Divide:

$12x^2 + 7xy - 12y^2$ by $3x + 4y$

Answer

$$(12x^2 + 7xy - 12y^2) \div (3x + 4y)$$

$$= (12x^2 + 16xy - 9xy - 12y^2) \div (3x + 4y)$$

$$= [4x(3x + 4y) - 3y(3x + 4y)] \div (3x + 4y)$$

$$= [(3x + 4y)(4x - 3y)] \div (3x + 4y)$$

$$= \frac{[(\cancel{3x + 4y})(4x - 3y)]}{(\cancel{3x + 4y})}$$

$$= 4x - 3y$$

$$\text{Hence, } (12x^2 + 7xy - 12y^2) \div (3x + 4y) = (4x - 3y)$$

Question 11(xii)

Divide:

$$x^6 - 8 \text{ by } x^2 - 2$$

Answer

$$(x^6 - 8) \div (x^2 - 2)$$

$$= (x^{2 \times 3} - 2^3) \div (x^2 - 2)$$

$$[\text{Using the formula } (a^3 - b^3) = (a - b)(a^2 + ab + b^2)]$$

$$= [(x^2 - 2)(x^4 + 2x^2 + 4)] \div (x^2 - 2)$$

$$= \frac{(\cancel{x^2 - 2})(x^4 + 2x^2 + 4)}{(\cancel{x^2 - 2})}$$

$$= (x^4 + 2x^2 + 4)$$

Hence, $(x^6 - 8) \div (x^2 - 2) = (x^4 + 2x^2 + 4)$

Question 11(xiii)

Divide:

$6x^3 - 13x^2 - 13x + 30$ by $2x^2 - x - 6$

Answer

Dividing $6x^3 - 13x^2 - 13x + 30$ by $2x^2 - x - 6$

$$\begin{array}{r}
 3x - 5 \\
 2x^2 - x - 6 \overline{) 6x^3 - 13x^2 - 13x + 30} \\
 \underline{6x^3 - 3x^2 - 18x} \\
 -10x^2 + 5x + 30 \\
 \underline{-10x^2 + 5x + 30} \\
 0
 \end{array}$$

Hence, $(6x^3 - 13x^2 - 13x + 30) \div (2x^2 - x - 6) = 3x - 5$

Question 11(xiv)

Divide:

$4a^2 + 12ab + 9b^2 - 25c^2$ by $2a + 3b + 5c$

Answer

Dividing $4a^2 + 12ab + 9b^2 - 25c^2$ by $2a + 3b + 5c$

$$\begin{array}{r}
 2a + 3b - 5c \\
 \hline
 2a + 3b + 5c \overline{) 4a^2 + 12ab + 9b^2 - 25c^2} \\
 \underline{4a^2 + 6ab} \\
 6ab - 10ac + 9b^2 - 25c^2 \\
 \underline{6ab} \\
 -10ac - 25c^2 - 15bc \\
 \underline{-10ac - 25c^2 - 15bc} \\
 0
 \end{array}$$

Hence, $(4a^2 + 12ab + 9b^2 - 25c^2) \div (2a + 3b + 5c) = 2a + 3b - 5c$

Question 11(xv)

Divide:

$$16 + 8x + x^6 - 8x^3 - 2x^4 + x^2 \text{ by } x + 4 - x^3$$

Answer

Dividing $16 + 8x + x^6 - 8x^3 - 2x^4 + x^2$ by $x + 4 - x^3$

$\Rightarrow$ Dividing $x^6 - 2x^4 - 8x^3 + x^2 + 8x + 16$ by $-x^3 + x + 4$

$$\begin{array}{r}
 -x^3 + x + 4 \\
 -x^3 + x + 4 \overline{) x^6 - 2x^4 - 8x^3 + x^2 + 8x + 16} \\
 \underline{x^6 - x^4 - 4x^3} \\
 -x^4 - 4x^3 + x^2 + 8x + 16 \\
 \underline{-x^4 + x^2 + 4x} \\
 -4x^3 + 4x + 16 \\
 \underline{-4x^3 - 4x + 16} \\
 + + 16 \\
 \times
 \end{array}$$

Hence, $(16 + 8x + x^6 - 8x^3 - 2x^4 + x^2) \div (x + 4 - x^3) = -x^3 + x + 4$

Question 12(i)

Find the quotient and the remainder (if any), when:

$a^3 - 5a^2 + 8a + 15$ is divided by $a + 1$.

Answer

Dividing $a^3 - 5a^2 + 8a + 15$ by $a + 1$

$$\begin{array}{r}
 a^2 - 6a + 14 \\
 a + 1 \overline{) a^3 - 5a^2 + 8a + 15} \\
 \underline{a^3 + a^2} \\
 -6a^2 + 8a + 15 \\
 \underline{-6a^2 - 6a} \\
 14a + 15 \\
 \underline{-14a + 14} \\
 1
 \end{array}$$

$$\text{Quotient} = a^2 - 6a + 14$$

$$\text{Remainder} = 1$$

Verification:

$$\text{Quotient} \times \text{Divisor} + \text{Remainder}$$

$$= (a^2 - 6a + 14) \times (a + 1) + 1$$

$$= a \times (a^2 - 6a + 14) + 1 \times (a^2 - 6a + 14) + 1$$

$$= a^{(1+2)} - 6a^{(1+1)} + 14a + a^2 - 6a + 14 + 1$$

$$= a^3 - 6a^2 + 14a + a^2 - 6a + 14 + 1$$

$$= a^3 + (-6a^2 + a^2) + (14a - 6a) + (14 + 1)$$

$$= a^3 - 5a^2 + 8a + 15$$

$$= \text{Dividend}$$

Question 12(ii)

Find the quotient and the remainder (if any), when:

$$3x^4 + 6x^3 - 6x^2 + 2x - 7 \text{ is divided by } x - 3.$$

Answer

$$\text{Dividing } 3x^4 + 6x^3 - 6x^2 + 2x - 7 \text{ by } x - 3$$

$$\begin{array}{r}
 3x^3 + 15x^2 + 39x + 119 \\
 x - 3 \overline{) 3x^4 + 6x^3 - 6x^2 + 2x - 7} \\
 \underline{3x^4 - 9x^3} \\
 15x^3 - 6x^2 + 2x - 7 \\
 \underline{+ 15x^3 - 45x^2} \\
 39x^2 + 2x - 7 \\
 \underline{+ 39x^2 - 117x} \\
 119x - 7 \\
 \underline{+ 119x - 357} \\
 350
 \end{array}$$

$$\text{Quotient} = 3x^3 + 15x^2 + 39x + 119$$

$$\text{Remainder} = 350$$

Verification:

$$\text{Quotient} \times \text{Divisor} + \text{Remainder}$$

$$= (3x^3 + 15x^2 + 39x + 119) \times (x - 3) + 350$$

$$= x \times (3x^3 + 15x^2 + 39x + 119) - 3 \times (3x^3 + 15x^2 + 39x + 119) + 350$$

$$= 3x^{(3+1)} + 15x^{(2+1)} + 39x^{(1+1)} + 119x - 9x^3 - 45x^2 - 117x - 357 + 350$$

$$= 3x^4 + 15x^3 + 39x^2 + 119x - 9x^3 - 45x^2 - 117x - 357 + 350$$

$$= 3x^4 + (15x^3 - 9x^3) + (39x^2 - 45x^2) + (119x - 117x) + (-357 + 350)$$

$$= 3x^4 + 6x^3 - 6x^2 + 2x - 7$$

= Dividend

Question 12(iii)

Find the quotient and the remainder (if any), when:

$6x^2 + x - 15$ is divided by $3x + 5$.

Answer

$$(6x^2 + x - 15) \div (3x + 5)$$

Dividing $6x^2 + x - 15$ by $3x + 5$

$$\begin{array}{r}
 2x - 3 \\
 3x + 5 \overline{) 6x^2 + x - 15} \\
 \underline{6x^2 + 10x} \\
 -9x - 15 \\
 \underline{-9x - 15} \\
 + + \\
 \hline
 \times
 \end{array}$$

$$\text{Quotient} = 2x - 3$$

$$\text{Remainder} = 0$$

Verification:

Quotient $\times$ Divisor + Remainder

$$= (2x - 3) \times (3x + 5) + 0$$

$$= 2x \times (3x + 5) - 3 \times (3x + 5)$$

$$= 6x^{(1+1)} + 10x - 9x - 15$$

$$= 6x^2 + (10x - 9x) - 15$$

$$= 6x^2 + x - 15$$

= Dividend

Question 13

The area of a rectangle is $x^3 - 8x^2 + 7$ and one of its sides is $x - 1$. Find the length of the adjacent side.

Answer

$$\text{Area of the rectangle} = x^3 - 8x^2 + 7$$

$$\text{One side} = (x - 1)$$

As we know the area of rectangle = One side x Other side

$$\text{Other side} = \text{Area of rectangle} \div \text{One side}$$

$$= (x^3 - 8x^2 + 7) \div (x - 1)$$

$$\begin{array}{r}
 x^2 - 7x - 7 \\
 x - 1 \overline{) x^3 - 8x^2 + 7} \\
 \underline{x^3 - x^2} \\
 -7x^2 + 7 \\
 \underline{-7x^2} \\
 7x + 7 \\
 \underline{-7x + 7} \\
 0
 \end{array}$$

Hence, length of adjacent side is $x^2 - 7x - 7$.

Exercise 11(C)

Question 1(i)

The value of $x - \overline{x - y}$ is:

1. $2x$

2. $-y$

3. y

4. $2x + y$

Answer

$$x - \overline{x - y}$$

$$= x - x + y$$

$$= y$$

Hence, option 3 is the correct option.**Question 1(ii)** $(5x - 4y) - (5y - 4x)$ is equal to:

1. $9(x - y)$

2. $x - y$

3. $(y - x)$

4. 0

Answer

$$(5x - 4y) - (5y - 4x)$$

$$= 5x - 4y - 5y + 4x$$

$$= (5x + 4x) + (-4y - 5y)$$

$$= 9x + (-9y)$$

$$= 9x - 9y$$

$$= 9(x - y)$$

Hence, option 1 is the correct option.

Question 1(iii)

$x(y - x - \overline{x - y})$ is equal to:

1. $2x(x - y)$

2. $2x(y - x)$

3. $2x(x + y)$

4. $-2x(y - x)$

Answer

$$x(y - x - \overline{x - y})$$

$$= x(y - x - x + y)$$

$$= x(2y - 2x)$$

$$= 2x(y - x)$$

Hence, option 2 is the correct option.

Question 1(iv)

$(2x - y) + \overline{2x - y}$ is equal to :

1. 0

2. $2y$

3. $4x$

4. $4x - 2y$

Answer

$$(2x - y) + \overline{2x - y}$$

$$= 2x - y + 2x - y$$

$$= 4x - 2y$$

Hence, option 4 is the correct option.**Question 1(v)** $x(y - z) + y(z - x) - z(y - x)$ is equal to:

1. 0

2. $xy + yz + zx$

3. $xy + yz - zx$

4. $xy - yz + zx$

Answer

$$x(y - z) + y(z - x) - z(y - x)$$

$$= xy - xz + yz - xy - yz + xz$$

$$= (xy - xy) + (-xz + xz) + (yz - yz)$$

$$= 0 + 0 + 0$$

Hence, option 1 is the correct option.

Question 2

$$a^2 - 2a + \{5a^2 - (3a - 4a^2)\}$$

Answer

$$a^2 - 2a + \{5a^2 - (3a - 4a^2)\}$$

$$= a^2 - 2a + \{5a^2 - 3a + 4a^2\}$$

$$= a^2 - 2a + \{(5a^2 + 4a^2) - 3a\}$$

$$= a^2 - 2a + \{9a^2 - 3a\}$$

$$= a^2 - 2a + 9a^2 - 3a$$

$$= (a^2 + 9a^2) + (-2a - 3a)$$

$$= 10a^2 + (-5a)$$

$$= 10a^2 - 5a$$

$$\text{Hence, } a^2 - 2a + \{5a^2 - (3a - 4a^2)\} = 10a^2 - 5a$$

Question 3

$$x - y - \{x - y - (x + y) - \overline{x - y}\}$$

Answer

$$\begin{aligned}
 & x - y - \{x - y - (x + y) - \overline{x - y}\} \\
 &= x - y - \{x - y - (x + y) - x + y\} \\
 &= x - y - \{\cancel{x} - \cancel{y} - (x + y) - \cancel{x} + \cancel{y}\} \\
 &= x - y - \{-(x + y)\} \\
 &= x - y - \{-x - y\} \\
 &= x - \cancel{y} + x + \cancel{y} \\
 &= 2x
 \end{aligned}$$

Hence, $x - y - \{x - y - (x + y) - \overline{x - y}\} = 2x$

Question 4

$$-3(1 - x^2) - 2\{x^2 - (3 - 2x^2)\}$$

Answer

$$\begin{aligned}
 & -3(1 - x^2) - 2\{x^2 - (3 - 2x^2)\} \\
 &= -3(1 - x^2) - 2\{x^2 - 3 + 2x^2\} \\
 &= -3(1 - x^2) - 2\{(x^2 + 2x^2) - 3\} \\
 &= -3(1 - x^2) - 2\{3x^2 - 3\} \\
 &= -3 + 3x^2 - 6x^2 + 6 \\
 &= (-3 + 6) + (3x^2 - 6x^2)
 \end{aligned}$$

$$= 3 + (-3x^2)$$

$$= 3 - 3x^2$$

$$\text{Hence, } -3(1 - x^2) - 2\{x^2 - (3 - 2x^2)\} = 3 - 3x^2$$

Question 5

$$2\{m - 3(n + \overline{m - 2n})\}$$

Answer

$$2\{m - 3(n + \overline{m - 2n})\}$$

$$= 2\{m - 3(n + m - 2n)\}$$

$$= 2\{m - 3((n - 2n) + m)\}$$

$$= 2\{m - 3(-n + m)\}$$

$$= 2\{m + 3n - 3m\}$$

$$= 2\{(m - 3m) + 3n\}$$

$$= 2\{-2m + 3n\}$$

$$= -4m + 6n$$

$$\text{Hence, } 2\{m - 3(n + \overline{m - 2n})\} = 6n - 4m$$

Question 6

$$3x - [3x - \{3x - (3x - \overline{3x - y})\}]$$

Answer

$$\begin{aligned}
& 3x - [3x - \{3x - (3x - \overline{3x - y})\}] \\
&= 3x - [3x - \{3x - (3x - 3x + y)\}] \\
&= 3x - [3x - \{3x - (y)\}] \\
&= 3x - [3x - \{3x - y\}] \\
&= 3x - [3x - 3x + y] \\
&= 3x - [y] \\
&= 3x - y
\end{aligned}$$

Hence, $3x - [3x - \{3x - (3x - \overline{3x - y})\}] = 3x - y$

Question 7

$$p^2x - 2\{px - 3x(x^2 - \overline{3a - x^2})\}$$

Answer

$$\begin{aligned}
& p^2x - 2\{px - 3x(x^2 - \overline{3a - x^2})\} \\
&= p^2x - 2\{px - 3x(x^2 - 3a + x^2)\} \\
&= p^2x - 2\{px - 3x((x^2 + x^2) - 3a)\} \\
&= p^2x - 2\{px - 3x(2x^2 - 3a)\} \\
&= p^2x - 2\{px - 6x^3 + 9ax\} \\
&= p^2x - 2px + 12x^3 - 18ax
\end{aligned}$$

Hence, $p^2x - 2\{px - 3x(x^2 - \overline{3a - x^2})\} = p^2x - 2px + 12x^3 - 18ax$

Question 8

$$2[6 + 4\{m - 6(7 - \overline{n + p}) + q\}]$$

Answer

$$\begin{aligned} & 2[6 + 4\{m - 6(7 - \overline{n + p}) + q\}] \\ &= 2[6 + 4\{m - 6(7 - n - p) + q\}] \\ &= 2[6 + 4\{m - 42 + 6n + 6p + q\}] \\ &= 2[6 + 4m - 168 + 24n + 24p + 4q] \\ &= 2[(6 - 168) + 4m + 24n + 24p + 4q] \\ &= 2[-162 + 4m + 24n + 24p + 4q] \\ &= -324 + 8m + 48n + 48p + 8q \end{aligned}$$

$$\text{Hence, } 2[6 + 4\{m - 6(7 - \overline{n + p}) + q\}] = 8m + 48n + 48p + 8q - 324$$

Question 9

$$a - [a - \overline{b + a} - \{a - (a - \overline{b - a})\}]$$

Answer

$$\begin{aligned} & a - [a - \overline{b + a} - \{a - (a - \overline{b - a})\}] \\ &= a - [a - \overline{b + a} - \{a - (a - b + a)\}] \\ &= a - [a - \overline{b + a} - \{a - (a + a - b)\}] \\ &= a - [a - \overline{b + a} - \{a - (2a - b)\}] \end{aligned}$$

$$= a - [a - \overline{b + a} - \{a - 2a + b\}]$$

$$= a - [a - \overline{b + a} - \{-a + b\}]$$

$$= a - [a - \overline{b + a} + a - b]$$

$$= a - [a - b - a + a - b]$$

$$= a - [(a - a + a) + (-b - b)]$$

$$= a - [a + (-2b)]$$

$$= a - [a - 2b]$$

$$= a - a + 2b$$

$$= 2b$$

$$\text{Hence, } a - [a - \overline{b + a} - \{a - (a - \overline{b - a})\}] = 2b$$

Question 10

$$3x - [4x - \overline{3x - 5y} - 3\{2x - (3x - \overline{2x - 3y})\}]$$

Answer

$$3x - [4x - \overline{3x - 5y} - 3\{2x - (3x - \overline{2x - 3y})\}]$$

$$= 3x - [4x - \overline{3x - 5y} - 3\{2x - (3x - 2x + 3y)\}]$$

$$= 3x - [4x - \overline{3x - 5y} - 3\{2x - (x + 3y)\}]$$

$$= 3x - [4x - \overline{3x - 5y} - 3\{2x - x - 3y\}]$$

$$= 3x - [4x - \overline{3x - 5y} - 3\{x - 3y\}]$$

$$= 3x - [4x - \overline{3x - 5y} - 3x + 9y]$$

$$= 3x - [4x - 3x + 5y - 3x + 9y]$$

$$= 3x - [(4x - 3x - 3x) + (5y + 9y)]$$

$$= 3x - [-2x + 14y]$$

$$= 3x + 2x - 14y$$

$$= 5x - 14y$$

$$\text{Hence, } 3x - [4x - \overline{3x - 5y} - 3\{2x - (3x - \overline{2x - 3y})\}] = 5x - 14y$$

Question 11

$$a^5 \div a^3 + 3a \times 2a$$

Answer

$$a^5 \div a^3 + 3a \times 2a$$

$$= a^{(5-3)} + 3a \times 2a$$

$$= a^2 + 3a \times 2a$$

$$= a^2 + 6a^{(1+1)}$$

$$= a^2 + 6a^2$$

$$= 7a^2$$

$$\text{Hence, } a^5 \div a^3 + 3a \times 2a = 7a^2$$

Question 12

$$x^5 \div (x^2 \times y^2) \times y^3$$

Answer

$$x^5 \div (x^2 \times y^2) \times y^3$$

$$= \frac{x^5}{x^2 y^2} \times y^3$$

$$= \frac{x^{(5-2)}}{y^2} \times y^3$$

$$= \frac{x^3}{y^2} \times y^3$$

$$= \frac{x^3 y^3}{y^2}$$

$$= x^3 y^{3-2}$$

$$= x^3 y^1$$

$$\text{Hence, } x^5 \div (x^2 \times y^2) \times y^3 = x^3 y$$

Question 13

$$(x^5 \div x^2) \times y^2 \times y^3$$

Answer

$$(x^5 \div x^2) \times y^2 \times y^3$$

$$= x^{(5-2)} \times y^2 \times y^3$$

$$= x^3 \times y^2 \times y^3$$

$$= x^3 \times y^{(2+3)}$$

$$= x^3 \times y^5$$

$$\text{Hence, } (x^5 \div x^2) \times y^2 \times y^3 = x^3 y^5$$

Question 14

$$(y^3 - 5y^2) \div y \times (y - 1)$$

Answer

$$(y^3 - 5y^2) \div y \times (y - 1)$$

$$= (y^3 \div y - 5y^2 \div y) \times (y - 1)$$

$$= (y^{(3-1)} - 5y^{(2-1)}) \times (y - 1)$$

$$= (y^2 - 5y^1) \times (y - 1)$$

$$= y^2 \times (y - 1) - 5y^1 \times (y - 1)$$

$$= y^{(2+1)} - y^2 - 5y^{(1+1)} - 5y$$

$$= y^3 - y^2 - 5y^2 - 5y$$

$$= y^3 - 6y^2 - 5y$$

$$\text{Hence, } (y^3 - 5y^2) \div y \times (y - 1) = y^3 - 6y^2 - 5y$$

Test Yourself

Question 1(i)

$(-18xy) - (-8xy)$ is equal to:

1. $10xy$
2. $-10xy$
3. $26xy$
4. $-26xy$

Answer

$$(-18xy) - (-8xy)$$

$$= -18xy + 8xy$$

$$= -10xy$$

Hence, option 2 is the correct option.

Question 1(ii)

$(9a + 7b - 6c) - (2a - 3b + 4c)$ is equal to:

1. $7x + 7b + 10c$
2. $7a + 10b - 10c$
3. $7a - 10b + 10c$
4. $7a - 10b - 10c$

Answer

$$(9a + 7b - 6c) - (2a - 3b + 4c)$$

$$= 9a + 7b - 6c - 2a + 3b - 4c$$

$$= (9a - 2a) + (7b + 3b) + (-6c - 4c)$$

$$= 7a + 10b + (-10c)$$

$$= 7a + 10b - 10c$$

Hence, option 2 is the correct option.

Question 1(iii)

$-81a^5b^4c^3 \div (-9a^2b^2c)$ is equal to:

1. $-9a^3b^2c^2$
2. $3a^3b^2c^2$

3. $9a^4b$

4. $9a^3b^2c^2$

Answer

$$\begin{aligned} & -81a^5b^4c^3 \div (-9a^2b^2c) \\ &= \frac{-81}{-9} a^{(5-2)}b^{(4-2)}c^{(3-1)} \\ &= 9a^3b^2c^2 \end{aligned}$$

Hence, option 4 is the correct option.**Question 1(iv)** $-(-pq - \overline{p^2 - pq})$ is equal to:

1. $2pq + p^2$

2. $-p^2$

3. p^2

4. none of these

Answer

$$\begin{aligned} & -(-pq - \overline{p^2 - pq}) \\ &= -(-pq - p^2 + pq) \\ &= -((-pq + pq) - p^2) \end{aligned}$$

$$= -(-p^2)$$

$$= p^2$$

Hence, option 3 is the correct option.

Question 1(v)

$x^3 - y^3 - x(y^2 + x^2 - z^2)$ is equal to:

1. $y^3 + xy^2 + xz^2$

2. $y^3 + xy^2 - xz^2$

3. $-y^3 + xy^2 - xz^2$

4. $-y^3 - xy^2 + xz^2$

Answer

$$x^3 - y^3 - x(y^2 + x^2 - z^2)$$

$$= x^3 - y^3 - x(y^2 + x^2 - z^2)$$

$$= x^3 - y^3 - xy^2 - x^3 + xz^2$$

$$= (x^3 - x^3) - y^3 - xy^2 + xz^2$$

$$= -y^3 - xy^2 + xz^2$$

Hence, option 4 is the correct option.

Question 1(vi)

Statement 1: The expression $2x^4 - 3x^2 + \frac{7}{x}$, $x \neq 0$ has no constant term.

Statement 2: In an algebraic expression in terms of one variable, the term(s) independent of the variable is called the constant.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

Given :

$$2x^4 - 3x^2 + \frac{7}{x}$$

A constant term is a term that does not contain the variable in this case, x .

Thus, the expression contains no constant term.

So, statement 1 is true.

In an algebraic expression in terms of one variable, the term(s) independent of the variable is called the constant.

This is a standard definition in algebra: a constant term is the part of the expression that does not change with the variable i.e., it is independent of the variable.

So, statement 2 is true.

Hence, option 1 is the correct option.

Question 1(vii)

Assertion (A) : $5x + y^2 - x^3$, $xy + yz + zx$, $x^2 - x + 1$ are all trinomials.

Reason (R) : An algebraic expression which contains three different terms is called a trinomials.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

An algebraic expression which contains three different terms is called a trinomial.

So, reason (R) is true.

Given, $5x + y^2 - x^3$, $xy + yz + zx$, $x^2 - x + 1$

All three expressions have three distinct terms.

So, assertion (A) is true and, reason (R) is the correct explanation of assertion (A).

Hence, option 1 is the correct option.

Question 1(viii)

Assertion (A) : Refer the following algebraic expressions in terms of one variable : $3x + 9$, $9 - \frac{3}{x^2}$, 100 .

Two of them are not polynomials in x .

Reason (R) : An algebraic expression is a polynomial if the degree of each term used in it is a whole number.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

An algebraic expression is a polynomial if the degree of each term used in it is a whole number.

So, reason (R) is true

A polynomial in one variable (say, x) must have:

- Only non-negative integers (i.e., whole numbers) as exponents of x .
- No negative or fractional powers.

So, reason (R) is true.

Given,

Algebraic expressions : $3x + 9$, $9 - \frac{3}{x^2}$, 100

$3x + 9$ and 100 (constant polynomial) both are polynomials, while $9 - \frac{3}{x^2}$ is not as on taking the variable in numerator ($9 - 3x^{-2}$) the power becomes negative.

So, assertion (A) is false.

Hence, option 4 is the correct option.

Question 1(ix)

Assertion (A) : $2xyz + 3x^2y$ is a cubic polynomial in three variables.

Reason (R) : An algebraic expression having two or more variables, the highest sum of the powers of all the variables in each term is taken as the degree of the polynomial, if this sum is a whole number.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

An algebraic expression having two or more variables, the highest sum of the powers of all the variables in each term taken as the degree of the polynomial if this sum is a whole number.

The maximum of these sums gives the degree. These powers must be whole numbers for it to be a valid polynomial.

So, reason (R) is true.

Given; $2xyz + 3x^2y$

Terms; $2xyz$, $3x^2y$

For term $2xyz$, degree = $1 + 1 + 1 = 3$

For term $3x^2y$, degree = $2 + 1 = 3$

The degree of a polynomial in multiple variables is the highest total degree among its terms.

Thus, the degree of this polynomial is 3.

Therefore, this is a cubic polynomial in three variables.

So, assertion (A) is true and, reason (R) is the correct explanation of assertion (A).

Hence, option 1 is the correct option.

Question 1(x)

Assertion (A) : $2024x^2yz$ is not a trinomial.

Reason (R) : For combining polynomials, the like terms of the given polynomials are combined together.

1. Both A and R are correct, and R is the correct explanation for A.

2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

A trinomial is an algebraic expression with exactly three terms.

Here, $2024x^2yz$ is just a single term — a monomial.

Thus, the statement that it's not a trinomial is correct.

So, assertion (A) is true.

For combining polynomials, the like terms of the given polynomials are combined together.

Like terms are terms with the same variables raised to the same powers.

When combining polynomials, you add or subtract like terms.

So, reason (R) is true but, reason (R) does not explain assertion(A).

Hence, option 2 is the correct option.

Question 2

In $\frac{5}{7}xy^2z^3$, write the coefficient of:

(i) 5

(ii) $\frac{5}{7}$

(iii) $5x$

(iv) xy^2

(v) z^3

(vi) xz^3

(vii) $5xy^2$

(viii) $\frac{1}{7}yz$

(ix) z

(x) yz^2

(xi) $5xyz$

Answer

(i) Coefficient = $\frac{1}{7}xy^2z^3$

(ii) Coefficient = xy^2z^3

(iii) Coefficient = $\frac{1}{7}y^2z^3$

(iv) Coefficient = $\frac{5}{7}z^3$

(v) Coefficient = $\frac{5}{7}xy^2$

(vi) Coefficient = $\frac{5}{7}y^2$

$$(vii) \text{ Coefficient} = \frac{1}{7} z^3$$

$$(viii) \text{ Coefficient} = 5 xyz^2$$

$$(ix) \text{ Coefficient} = \frac{5}{7} xy^2z^2$$

$$(x) \text{ Coefficient} = \frac{5}{7} xyz$$

$$(xi) \text{ Coefficient} = \frac{1}{7} yz^2$$

Question 3(i)

In each polynomial, given below, separate the like terms:

$$3xy, -4yx^2, 2xy^2, 2.5x^2y, -8yx, -3.2y^2x \text{ and } x^2y$$

Answer

Like terms are those terms which have the same variable raised to the same power.

Hence, $3xy, -8yx$; $2y^2x, -3.2y^2x$; $-4x^2y, 2.5x^2y$ and x^2y are like terms.

Question 3(ii)

In each polynomial, given below, separate the like terms:

$$y^2z^3, xy^2z^3, -5x^2yz, -4y^2z^3, -8xz^3y^2, 3x^2yz \text{ and } 2z^3y^2$$

Answer

Like terms are those terms which have the same variable raised to the same power.

Hence, y^2z^3 , $-4y^2z^3$ and $2z^3y^2$; xy^2z^3 , $-8xz^3y^2$; $-5x^2yz$, $3x^2yz$ are like terms.

Question 4

The sides of a triangle are $x^2 - 3xy + 8$, $4x^2 + 5xy - 3$ and $6 - 3x^2 + 4xy$. Find its perimeter.

Answer

The sides of a triangle = $(x^2 - 3xy + 8)$, $(4x^2 + 5xy - 3)$ and $(6 - 3x^2 + 4xy)$

Perimeter of triangle = sum of all its side.

$$(x^2 - 3xy + 8) + (4x^2 + 5xy - 3) + (6 - 3x^2 + 4xy)$$

$$= x^2 - 3xy + 8 + 4x^2 + 5xy - 3 + 6 - 3x^2 + 4xy$$

$$= (x^2 + 4x^2 - 3x^2) + (-3xy + 5xy + 4xy) + (-3 + 6 + 8)$$

$$= 2x^2 + 6xy + 11$$

Hence, perimeter of triangle = $2x^2 + 6xy + 11$

Question 5

The perimeter of a triangle is $8y^2 - 9y + 4$ and its two sides are $3y^2 - 5y$ and $4y^2 + 12$. Find its third side.

Answer

The perimeter of a triangle = $8y^2 - 9y + 4$

Two sides are $3y^2 - 5y$ and $4y^2 + 12$

Let the third side be S.

As we know that, perimeter of triangle = sum of all its side.

$$8y^2 - 9y + 4 = (3y^2 - 5y) + (4y^2 + 12) + S$$

$$\Rightarrow 8y^2 - 9y + 4 = 3y^2 - 5y + 4y^2 + 12 + S$$

$$\Rightarrow 8y^2 - 9y + 4 = (3y^2 + 4y^2) - 5y + 12 + S$$

$$\Rightarrow 8y^2 - 9y + 4 = 7y^2 - 5y + 12 + S$$

$$\Rightarrow S = (8y^2 - 9y + 4) - (7y^2 - 5y + 12)$$

$$\Rightarrow S = 8y^2 - 9y + 4 - 7y^2 + 5y - 12$$

$$\Rightarrow S = (8y^2 - 7y^2) + (-9y + 5y) + (4 - 12)$$

$$\Rightarrow S = 1y^2 - 4y - 8$$

Hence, the third side of triangle = $(y^2 - 4y - 8)$

Question 6

The two adjacent sides of a rectangle are $2x^2 - 5xy + 3z^2$ and $4xy - x^2 - z^2$. Find its perimeter.

Answer

The two adjacent sides of a rectangle = $(2x^2 - 5xy + 3z^2)$ and $(4xy - x^2 - z^2)$

As we know the perimeter of rectangle = $2 \times (l + b)$

$$= 2 \times [(2x^2 - 5xy + 3z^2) + (4xy - x^2 - z^2)]$$

$$= 2 \times [2x^2 - 5xy + 3z^2 + 4xy - x^2 - z^2]$$

$$= 2 \times [(2x^2 - x^2) + (-5xy + 4xy) + (3z^2 - z^2)]$$

$$= 2 \times [x^2 - 1xy + 2z^2]$$

$$= 2 \times x^2 - 2 \times 1xy + 2 \times 2z^2$$

$$= 2x^2 - 2xy + 4z^2$$

Hence, the perimeter of rectangle = $2x^2 - 2xy + 4z^2$

Question 7

What must be subtracted from $19x^4 + 2x^3 + 30x - 37$ to get $8x^4 + 22x^3 - 7x - 60$?

Answer

Let A be the number that should be subtracted from $19x^4 + 2x^3 + 30x - 37$ to get

$$8x^4 + 22x^3 - 7x - 60$$

$$(19x^4 + 2x^3 + 30x - 37) - A = (8x^4 + 22x^3 - 7x - 60)$$

$$\Rightarrow (19x^4 + 2x^3 + 30x - 37) - (8x^4 + 22x^3 - 7x - 60) = A$$

$$\Rightarrow A = 19x^4 + 2x^3 + 30x - 37 - 8x^4 - 22x^3 + 7x + 60$$

$$\Rightarrow A = (19x^4 - 8x^4) + (2x^3 - 22x^3) + (30x + 7x) + (-37 + 60)$$

$$\Rightarrow A = 11x^4 - 20x^3 + 37x + 23$$

Hence, the number is $11x^4 - 20x^3 + 37x + 23$.

Question 8

How much smaller is $15x - 18y + 19z$ than $22x - 20y - 13z + 26$?

Answer

$$(22x - 20y - 13z + 26) - (15x - 18y + 19z)$$

$$= 22x - 20y - 13z + 26 - 15x + 18y - 19z$$

$$= (22x - 15x) + (-20y + 18y) + (-13z - 19z) + 26$$

$$= 7x - 2y - 32z + 26$$

Hence, $(22x - 20y - 13z + 26)$ is less than $(15x - 18y + 19z)$ by $(7x - 2y - 32z + 26)$

Question 9

How much bigger is $5x^2y^2 - 18xy^2 - 10x^2y$ than $-5x^2 + 6x^2y - 7xy$?

Answer

$$(5x^2y^2 - 18xy^2 - 10x^2y) - (-5x^2 + 6x^2y - 7xy)$$

$$= 5x^2y^2 - 18xy^2 - 10x^2y + 5x^2 - 6x^2y + 7xy$$

$$= 5x^2y^2 - 18xy^2 + (-10x^2y - 6x^2y) + 5x^2 + 7xy$$

$$= 5x^2y^2 - 18xy^2 - 16x^2y + 5x^2 + 7xy$$

Hence, $(5x^2y^2 - 18xy^2 - 10x^2y)$ is bigger than $(-5x^2 + 6x^2y - 7xy)$ by $(5x^2y^2 - 18xy^2 - 16x^2y + 5x^2 + 7xy)$

Question 10

If $x = 2$ and $y = 1$; find the value of $(-4x^2y^3) \times (-5x^2y^5)$.

Answer

$$(-4x^2y^3) \times (-5x^2y^5)$$

$$= (-4) \times (-5)x^{(2+2)}y^{(3+5)}$$

$$= 20 x^4y^8$$

When $x = 2$ and $y = 1$,

$$= 20 \times 2^4 \times 1^8$$

$$= 20 \times 16 \times 1$$

$$= 320$$

Hence, if $x = 2$ and $y = 1$; the value of $(-4x^2y^3) \times (-5x^2y^5) = 320$.

Question 11(i)

Evaluate:

$(3x - 2)(x + 5)$ for $x = 2$.

Answer

$$(3x - 2)(x + 5)$$

$$= 3x(x + 5) - 2(x + 5)$$

$$= 3x^{(1+1)} + 15x - 2x - 10$$

$$= 3x^2 + (15x - 2x) - 10$$

$$= 3x^2 + 13x - 10$$

For $x = 2$

$$3 \times 2^2 + 13 \times 2 - 10$$

$$= 3 \times 4 + 26 - 10$$

$$= 12 + 26 - 10$$

$$= 38 - 10$$

$$= 28$$

Hence, for $x = 2$, the value of $(3x - 2)(x + 5)$ is 28.

Question 11(ii)

Evaluate:

$(2x - 5y)(2x + 3y)$ for $x = 2$ and $y = 3$.

Answer

$$\begin{aligned}& (2x - 5y)(2x + 3y) \\&= 2x(2x + 3y) - 5y(2x + 3y) \\&= 4x^{(1+1)} + 6xy - 10xy - 15y^{(1+1)} \\&= 4x^2 + (6xy - 10xy) - 15y^2 \\&= 4x^2 - 4xy - 15y^2\end{aligned}$$

For $x = 2$, $y = 3$

$$\begin{aligned}& 4 \times 2^2 - 4 \times 2 \times 3 - 15 \times 3^2 \\&= 4 \times 4 - 4 \times 2 \times 3 - 15 \times 9 \\&= 16 - 24 - 135 \\&= -143\end{aligned}$$

Hence, for $x = 2$, $y = 3$, the value $(2x - 5y)(2x + 3y) = -143$.

Question 11(iii)

Evaluate:

$$xz(x^2 + y^2) \text{ for } x = 2, y = 1 \text{ and } z = 1.$$

Answer

$$\begin{aligned}& xz(x^2 + y^2) \\&= x^{(2+1)}z + xzy^2\end{aligned}$$

$$= x^3z + xzy^2$$

For $x = 2$, $y = 1$ and $z = 1$

$$2^3 \times 1 + 2 \times 1 \times 1^2$$

$$= 8 \times 1 + 2 \times 1 \times 1$$

$$= 8 + 2$$

$$= 10$$

Hence, for $x = 2$, $y = 1$ and $z = 1$, the value of $xz(x^2 + y^2)$ is 10.

Question 12(i)

Evaluate:

$$x(x - 5) + 2 \text{ for } x = 1.$$

Answer

$$x(x - 5) + 2$$

$$= x^2 - 5x + 2$$

$$\text{For } x = 1$$

$$= 1^2 - 5 \times 1 + 2$$

$$= 1 - 5 + 2$$

$$= -2$$

Hence, for $x = 1$, the value of $x(x - 5) + 2$ is -2.

Question 12(ii)

Evaluate:

$$xy^2(x - 5y) + 1 \text{ for } x = 2 \text{ and } y = 1.$$

Answer

$$xy^2(x - 5y) + 1$$

$$= x^{(1+1)}y^2 - 5xy^{(2+1)} + 1$$

$$= x^2y^2 - 5xy^3 + 1$$

For $x = 2$ and $y = 1$,

$$2^2 \times 1^2 - 5 \times 2 \times 1^3 + 1$$

$$= 4 \times 1 - 5 \times 2 \times 1 + 1$$

$$= 4 - 10 + 1$$

$$= -5$$

Hence, for $x = 2$ and $y = 1$, the value of $[xy^2(x - 5y) + 1]$ is -5 .**Question 12(iii)**

Evaluate:

$$2x(3x - 5) - 5(x - 2) - 18 \text{ for } x = 2.$$

Answer

$$2x(3x - 5) - 5(x - 2) - 18$$

$$= 6x^{(1+1)} - 10x - 5x + 10 - 18$$

$$= 6x^2 + (-10x - 5x) + (10 - 18)$$

$$= 6x^2 - 15x - 8$$

For $x = 2$,

$$= 6 \times 2^2 - 15 \times 2 - 8$$

$$= 6 \times 4 - 15 \times 2 - 8$$

$$= 24 - 30 - 8$$

$$= -14$$

Hence, for $x = 2$, the value of $[2x(3x - 5) - 5(x - 2) - 18]$ is -14 .

Question 13

Multiply and then verify:

$-3x^2y^2$ and $(x - 2y)$ for $x = 1$ and $y = 2$.

Answer

$$(-3x^2y^2) \times (x - 2y)$$

$$= (-3x^2y^2) \times x - (-3x^2y^2) \times 2y$$

$$= (-3x^{(2+1)}y^2) - (-6x^2y^{(2+1)})$$

$$= -3x^3y^2 + 6x^2y^3$$

For $x = 1$ and $y = 2$

$$(-3x^2y^2) \times (x - 2y) = -3x^3y^2 + 6x^2y^3$$

Taking LHS:

$$(-3x^2y^2) \times (x - 2y)$$

$$= (-3 \times 1^2 \times 2^2) \times (1 - 2 \times 2)$$

$$= (-3 \times 1 \times 4) \times (1 - 4)$$

$$= (-12) \times (-3)$$

$$= 36$$

Now, taking RHS:

$$-3x^3y^2 + 6x^2y^3$$

$$= -3 \times 1^3 \times 2^2 + 6 \times 1^2 \times 2^3$$

$$= -3 \times 1 \times 4 + 6 \times 1 \times 8$$

$$= -12 + 48$$

$$= 36$$

Hence, LHS = RHS.

Question 14(i)

Multiply:

$$2x^2 - 4x + 5 \text{ by } x^2 + 3x - 7$$

Answer

$$(2x^2 - 4x + 5) \times (x^2 + 3x - 7)$$

$$= 2x^2 \times (x^2 + 3x - 7) - 4x \times (x^2 + 3x - 7) + 5 \times (x^2 + 3x - 7)$$

$$= 2x^{(2+2)} + 6x^{(2+1)} - 14x^2 - 4x^{(1+2)} - 12x^{(1+1)} + 28x + 5x^2 + 15x - 35$$

$$= 2x^4 + 6x^3 - 14x^2 - 4x^3 - 12x^2 + 28x + 5x^2 + 15x - 35$$

$$= 2x^4 + (6x^3 - 4x^3) + (-14x^2 - 12x^2 + 5x^2) + (28x + 15x) - 35$$

$$= 2x^4 + 2x^3 - 21x^2 + 43x - 35$$

$$\text{Hence, } (2x^2 - 4x + 5) \times (x^2 + 3x - 7) = 2x^4 + 2x^3 - 21x^2 + 43x - 35$$

Question 14(ii)

Multiply:

$$(ab - 1)(3 - 2ab)$$

Answer

$$(ab - 1)(3 - 2ab)$$

$$= ab(3 - 2ab) - 1(3 - 2ab)$$

$$= 3ab - 2a^{(1+1)}b^{(1+1)} - 3 + 2ab$$

$$= 3ab - 2a^2b^2 - 3 + 2ab$$

$$= (3ab + 2ab) - 2a^2b^2 - 3$$

$$= 5ab - 2a^2b^2 - 3$$

$$\text{Hence, } (ab - 1)(3 - 2ab) = 5ab - 2a^2b^2 - 3$$

Question 15

Simplify:

$$(5 - x)(6 - 5x)(2 - x).$$

Answer

$$(5 - x)(6 - 5x)(2 - x)$$

$$= [5(6 - 5x) - x(6 - 5x)](2 - x)$$

$$= [30 - 25x - 6x + 5x^{(1+1)}](2 - x)$$

$$= [30 + (-25x - 6x) + 5x^2](2 - x)$$

$$= [30 - 31x + 5x^2](2 - x)$$

$$= 2(30 - 31x + 5x^2) - x(30 - 31x + 5x^2)$$

$$= 60 - 62x + 10x^2 - 30x + 31x^{(1+1)} - 5x^{(2+1)}$$

$$= 60 - 62x + 10x^2 - 30x + 31x^2 - 5x^3$$

$$= 60 + (-62x - 30x) + (10x^2 + 31x^2) - 5x^3$$

$$= 60 - 92x + 41x^2 - 5x^3$$

Hence, $(5 - x)(6 - 5x)(2 - x) = 60 - 92x + 41x^2 - 5x^3$

Question 16

The product of two numbers is $16x^4 - 1$. If one number is $2x - 1$, find the other.

Answer

The product = $(16x^4 - 1)$

One number = $(2x - 1)$

Let the other number be A.

$$(2x - 1) \times A = (16x^4 - 1)$$

$$\Rightarrow A = \frac{(16x^4 - 1)}{(2x - 1)}$$

$$\Rightarrow A = \frac{(4x^2 - 1)(4x^2 + 1)}{(2x - 1)}$$

$$\Rightarrow A = \frac{(2x - 1)(2x + 1)(4x^2 + 1)}{(2 - 1)}$$

$$\Rightarrow A = \frac{\cancel{(2x - 1)}(2x + 1)(4x^2 + 1)}{\cancel{(2x - 1)}}$$

$$\Rightarrow A = (2x + 1)(4x^2 + 1)$$

$$\Rightarrow A = 2x(4x^2 + 1) + 1(4x^2 + 1)$$

$$\Rightarrow A = 8x^{(2+1)} + 2x + 4x^2 + 1$$

$$\Rightarrow A = 8x^3 + 4x^2 + 2x + 1$$

Hence, the other number is $8x^3 + 4x^2 + 2x + 1$.

Question 17

Divide $x^6 - y^6$ by the product of $x^2 + xy + y^2$ and $x - y$.

Answer

The product of $(x^2 + xy + y^2)$ and $(x - y)$

$$= (x^2 + xy + y^2) \times (x - y)$$

$$= x \times (x^2 + xy + y^2) - y \times (x^2 + xy + y^2)$$

$$= x^{(1+2)} + x^{(1+1)}y + xy^2 - x^2y - xy^{(1+1)} - y^{(1+2)}$$

$$= x^3 + x^2y + xy^2 - x^2y - xy^2 - y^3$$

$$= x^3 + (x^2y - x^2y) + (xy^2 - xy^2) - y^3$$

$$= x^3 - y^3$$

$$\text{Now, } (x^6 - y^6) \div (x^3 - y^3)$$

$$\begin{array}{r}
 x^3 + y^3 \\
 x^3 - y^3 \overline{) x^6 - y^6} \\
 \underline{x^6 } \\
 x^3 y^3 - y^6 \\
 \underline{ x^3 y^3 - y^6} \\
 y^6 \\
 \underline{ y^6} \\
 0
 \end{array}$$

Hence, the answer is $(x^3 + y^3)$

Question 18

$$3a \times [8b \div 4 - 6\{a - (5a - \overline{3b - 2a})\}]$$

Answer

$$\begin{aligned}
 & 3a \times [8b \div 4 - 6\{a - (5a - \overline{3b - 2a})\}] \\
 &= 3a \times [8b \div 4 - 6\{a - (5a - 3b + 2a)\}] \\
 &= 3a \times [8b \div 4 - 6\{a - (5a + 2a - 3b)\}] \\
 &= 3a \times [8b \div 4 - 6\{a - (7a - 3b)\}] \\
 &= 3a \times [8b \div 4 - 6\{a - 7a + 3b\}] \\
 &= 3a \times [8b \div 4 - 6\{-6a + 3b\}] \\
 &= 3a \times [8b \div 4 + 36a - 18b] \\
 &= 3a \times [2b + 36a - 18b] \\
 &= 3a \times [2b - 18b + 36a] \\
 &= 3a \times [-16b + 36a]
 \end{aligned}$$

$$= -48ab + 108a^2$$

$$\text{Hence, } 3a \times [8b \div 4 - 6\{a - (5a - \overline{3b - 2a})\}] = 108a^2 - 48ab$$

Question 19

$$7x + 4\{x^2 \div (5x \div 10)\} - 3\{2 - x^3 \div (3x^2 \div x)\}$$

Answer

$$7x + 4\{x^2 \div (5x \div 10)\} - 3\{2 - x^3 \div (3x^2 \div x)\}$$

$$= 7x + 4\{x^2 \div (5x \div 10)\} - 3\{2 - x^3 \div 3x\}$$

$$= 7x + 4\{x^2 \div (5x \div 10)\} - 3\{2 - \frac{x^2}{3}\}$$

$$= 7x + 4\{x^2 \div (5x \div 10)\} - 6 + \frac{3x^2}{3}$$

$$= 7x + 4\{x^2 \div \frac{5x}{10}\} - 6 + x^2$$

$$= 7x + 4\{x^2 \div \frac{x}{2}\} - 6 + x^2$$

$$= 7x + 4\{\frac{2x^2}{x}\} - 6 + x^2$$

$$= 7x + 4 \times 2x - 6 + x^2$$

$$= 7x + 8x - 6 + x^2$$

$$= x^2 + 15x - 6$$

Hence, the answer is $x^2 + 15x - 6$

[Prev](#)
[← Direct and Inverse Variations](#)

[Next](#)
[Identities →](#)

Discover more

Social Sciences



Teaching & Classroom Resources



Textbooks



ICSE/ISC TEXTBOOK SOLUTIONS

[Class 6 ICSE Textbook Solutions](#)

[Class 7 ICSE Textbook Solutions](#)

[Class 8 ICSE Textbook Solutions](#)

[Class 9 ICSE Textbook Solutions](#)

[Class 10 ICSE Textbook Solutions](#)

[Class 11 ISC Textbook Solutions](#)

[Class 12 ISC Textbook Solutions](#)

[Selina Solutions for Class 10, 9, 8, 7, 6](#)

CBSE TEXTBOOK SOLUTIONS

[Class 8 CBSE Textbook Solutions](#)

[Class 9 CBSE Textbook Solutions](#)

[Class 10 CBSE Textbook Solutions](#)

[Class 11 CBSE Textbook Solutions](#)

[Class 12 CBSE Textbook Solutions](#)

ICSE/ISC/CBSE SOLVED QUESTION PAPERS

[ICSE Previous Year Question Papers Solved Class 10](#)

[ICSE Previous Year Question Papers Solved Class 12](#)

COMPANY

[Pricing](#)

[About Us](#)

[Contact Us](#)

[Privacy Policy](#)

[Terms of Service](#)

Morning Star Solutions for Class 10, 9

ICSE/ISC PRACTICE TESTS

Class 8 ICSE Practice Tests

Class 9 ICSE Practice Tests

Class 10 ICSE Practice Tests

CBSE Previous Year Question Papers Solved Class 10

CBSE Previous Year Question Papers Solved Class 12

ICSE Competency Focussed Practice Questions Solved Class 10

CBSE PRACTICE TESTS

Class 10 CBSE Practice Tests

Class 12 CBSE Practice Tests



Copyright © KnowledgeBoat 2026